



Entegris Commands Document

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DESCRIPTION:

This document details the serial commands, status, and return codes for the IntelliGen RULV, ULV, LV, MV, HMV, and HV20 generation dispense systems.

This document is auto generated directly from the source code of MMI/FIRMWARE.

There is an [appendix](#) included that provides more detailed information about some subjects.

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Commands

Commands (Alphabetical)

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Commands (Numeric)

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UtilityStatus	106	
Get Real Time Status for Cyberspace	110	
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Save Reference	156	
Job Information		157
DC Status	158	

Vent only start	160	
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Utility: Set motor positions	183	184
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Self Test: Fill Motor Strength	203	204
Self Test: Dry Dis Motor Strength	205	206
Self Test: Pressure Sensor Accuracy	207	208
Start Auto_Rec	209	
Self Test: Pump Leak Test	211	212
Cancel Self Test	213	
Self Test: Air Detect	214	215
Self Test: Pressure Rezero	216	217
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SOFTWARE_RESET: Software Reset

This command will restart the pump. Similar to a power-on condition.

Command Code: 2 SOFTWARE_RESET	
Send Data	None
Receive Data	None
Access Level	OPERATOR

WRITE_NOTE: Write user note Size: 500 bytes

This command is used to enter a user message into the log.

Command Code: 3 WRITE_NOTE	
Send Data	500 bytes data as defined below
Receive Data	None
Access Level	NONE

Write user note Structure			Size: 500 bytes
Type	Field	Description	
0 STRING	Message[500]	User message Default: 0	

CLEAR_ERRORS: Clear errors

This command clears the pending warnings and errors. Depending on the type of error, it may also re-initialize the pump.

Command Code: 4 CLEAR_ERRORS	
Send Data	None
Receive Data	None
Access Level	OPERATOR

START_MAINTENANCE: Go into Maintenance mode

This command will start the put the pump into maintenance mode.

Command Code: 6 START_MAINTENANCE	
Send Data	None
Receive Data	None
Access Level	ENGINEER

END_MAINTENANCE: End Maintenance mode

This command will remove the system from maintenance mode.

Command Code: 8 END_MAINTENANCE	
Send Data	<i>None</i>
Receive Data	<i>None</i>
Access Level	ENGINEER

OFFLINE_MODE: Go into offline mode Size: 2 bytes

This command will put the system in and out of the offline mode.

Command Code: 98 OFFLINE_MODE	
Send Data	2 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Go into offline mode Structure			Size: 2 bytes
Type	Field	Description	
0 UINT16	Command	Go Offline = 1, Go Online = 0 Default: 0	

TIME: Time Now Size: 4 bytes

Gets and sets the Real Time Clock value. The value is the number of seconds since 1970 (UNIX Epoch Time) in UTC timezone.

Command Code: 10 TIME	
Send Data	None
Receive Data	4 bytes data as defined below
Access Level	ENGINEER

Command Code: 11 WRITE TIME	
Send Data	4 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Time Now Structure		Size: 4 bytes
Type	Field	Description
0 ULONG	Time	UTC seconds from 1 Jan 1970 Min: 1381247363 Default: 0

SET_SYSTEM_VALVES: Set System valves Size: 2 bytes

This command will set the status of the system valves. See the [Appendix](#) for more detailed information about this command.

Command Code: 12 SET_SYSTEM_VALVES	
Send Data	2 bytes data as defined below
Receive Data	None
Access Level	OPERATOR

Set System valves Structure			Size: 2 bytes
Type	Field	Description	
0 UINT16	Valves	Bits for valves to be set Default: 0	

RESET_COUNTERS: Reset Counters Size: 4 bytes

This command is used to reset the resettable cycle counters (see CYCLECOUNTERS).

Command Code: 14 RESET_COUNTERS	
Send Data	4 bytes data as defined below
Receive Data	None
Access Level	OPERATOR

Reset Counters Structure			Size: 4 bytes
Type	Field	Description	
0 ULONG	ResetCounters	Which counters to reset: A=0 or B=1 Select which cycle counters to reset: A=0 or B=1 Max: 1 Default: 0	

GET_SVIDS: Get SVIDS

This command will accept a list of USHORT SVID values and return the appropriate data. Note: The size of the data depends on the SVID type. For strings the first USHORT is the length of the string.

Command Code: 21 GET_SVIDS	
Send Data	List of SVIDs (see SVIDs Appendix)
Receive Data	List of SVID data (see SVIDs Appendix)
Access Level	NONE

GET_ALARMS: Get Alarms Size: 8 bytes

Used to get a number of alarm records. See the [Appendix](#) for more detailed information about this command.

Command Code: 24 GET_ALARMS	
Send Data	8 bytes data as defined below
Receive Data	Alarms data (See Alarms Appendix)
Access Level	NONE

Get Alarms Structure			Size: 8 bytes
Type	Field	Description	
0 ULONG	AlarmNumber	First alarm to get (0 means most recent) Default: 0	
4 UINT16	Count	How many to get Min: 1 Max: 25 Default: 0	
6 UINT16	Reserved	Reserved Default: 0	

READ_RT_TRACES: Read Real Time Traces Size: 4 bytes

This is typically used every second to poll the pump for greater than 1 second of data. The return data includes a sample alignment number that is used to align the data correctly and discard the unused data. This allows real time display to overcome communication delays and retries. See the [Appendix](#) for more detailed information about this command.

Command Code: 26 READ_RT_TRACES	
Send Data	4 bytes data as defined below
Receive Data	Interleaved Trace data (See RT_TRACES Appendix)
Access Level	NONE

Read Real Time Traces Structure			Size: 4 bytes
Type	Field	Description	
0	UINT16	Points	How many points to get Min: 1 Max: 50 Default: 0
2	UINT16	Traces	Which Traces See the Appendix for more information. Min: 1 Max: ST_TRACE_ALL Default: 0

SERIALNUMBERS: Serial Numbers Size: 296 bytes

Reads the serial numbers for the system and some of its components.

Command Code: 30 SERIALNUMBERS	
Send Data	None
Receive Data	296 bytes data as defined below
Access Level	None

Serial Numbers Structure			Size: 296 bytes
Type	Field	Description	
0 STRING	Pump[24]	Pump Serial Number Default: "(None)"	
24 STRING	FillMotor[24]	Fill Motor Serial Number Default: "(None)"	
48 STRING	DispenseMotor[24]	Dispense Motor Serial Number Default: "(None)"	
72 STRING	FillPressure[24]	Fill Pressure Serial Number Default: "(None)"	
96 STRING	DispensePressure[24]	Dispense Pressure Serial Number Default: "(None)"	
120 STRING	MainBoard[24]	Main PCB Serial Number Default: "(None)"	
144 STRING	InternalBoard[24]	Internal PCB Serial Number Default: "(None)"	
168 STRING	InterfaceBoard[24]	Interface PCB Serial Number Default: "(None)"	
192 STRING	Reserved1[24]	Reserved Default: "RESERVED"	
216 STRING	Reserved2[24]	Reserved Default: "RESERVED"	
240 STRING	Reserved3[24]	Reserved Default: "RESERVED"	
264 STRING	Reserved4[24]	Reserved Default: "RESERVED"	
288 ULONG	Time	Time Written Default: 0	
292 ULONG	Cycle	Cycle when written Default: 0	

GET_PROFILE_TRACE: Get Profile Trace Data Size: 12 bytes

Get a piece of a profile trace. The profile can be a recently dispensed profile or a saved reference profile. See the [Appendix](#) for more detailed information about this command.

Command Code: 32 GET_PROFILE_TRACE	
Send Data	12 bytes data as defined below
Receive Data	Trace Data (See Profiles Appendix)
Access Level	NONE

Get Profile Trace Data Structure			Size: 12 bytes
Type	Field	Description	
0 ULONG	Cycle	Cycle Number Min: 1 Default: 0	
4 UINT16	Trace	Which Trace See the Appendix for more information. Max: 10 Default: 0	
6 UINT16	Size	How many USHORT Max: 500 Default: 0	
8 ULONG	Start	Starting USHORT Default: 0	

GET_PROFILE_TRACE_INFO: Get profile trace sizes Size: 76 bytes

Information about a recently dispensed profile or a saved reference profile. See the [Appendix](#) for more detailed information about this command.

Command Code: 33 GET_PROFILE_TRACE_INFO	
Send Data	4 bytes for Cycle
Receive Data	76 bytes data as defined below
Access Level	NONE

Get profile trace sizes Structure			Size: 76 bytes
Type	Field	Description	
0 ULONG	Cycle	Cycle Number Default: 0	
4 UINT16	Type_Encoding	Low Byte: (Encoding 0=None, 1=ERL16) Upper Byte: (Profile Type) Default: 0	
6 UINT16	SampleInterval	Time between samples Default: 0	
8 ULONG	Time	Time of dispense (trigger) Default: 0	
12 ULONG	StartCycle	Sample number for the start of the dispense Default: 0 Units: ms	
16 ULONG	DispenseEnd	Sample number for end of the dispense Default: 0	
20 ULONG	EndFiltration	Sample number for end of filtration Default: 0	
24 ULONG	EndCycle	Sample number for end of the cycle Default: 0	
28 ULONG	FillEnd	Sample number for end of the dispense Default: 0	
32 ULONG	UnencodedSize	Uncompressed size Default: 0	
36 ULONG	EncodedSize[10]	Compressed size Default: 0	

VERSION: Get Version String Size: 104 bytes

Gets version and model numbers strings for the system firmware.

Command Code: 52 VERSION	
Send Data	None
Receive Data	104 bytes data as defined below
Access Level	NONE

Get Version String Structure			Size: 104 bytes
Type	Field	Description	
0 STRING	reserved1[16]	reserved Default: 0	
16 STRING	reserved2[16]	reserved Default: 0	
32 STRING	Model[24]	Model Name Default: 0	
56 STRING	Version[48]	Firmware Version String Default: 0	

FIRMWAREINFO: Get Firmware Information Size: 16 bytes

Gets version and other information about the firmware including the product code. See the [Appendix](#) for more detailed information about this command.

Command Code: 54 FIRMWAREINFO	
Send Data	None
Receive Data	16 bytes data as defined below
Access Level	NONE

Get Firmware Information Structure			Size: 16 bytes
Type	Field	Description	
0	UINT16	ProductQualifier	Qualifer for ProductCode to be valid Default: 1234
2	UINT16	Comms_Version	Communications protocol version Default: 0
4	UINT16	MajorVersion	Major Version Number Default: 1011
6	UINT16	ModuleType	Interface Module Type Default: 0
8	UINT16	MinorVersion	Firmware Minor Version Default: 1536
10	UINT16	ProductCode	Product ID Code Default: 0
12	UINT16	MapSize	NV Map size (deprecated) Default: 0
14	UINT16	CRC	CRC (deprecated) Default: 0

CYCLECOUNTERS: Get Cycle Counters Size: 112 bytes

Reads all of the resettable and non-resettable cycle counters.

Command Code: 60 CYCLECOUNTERS	
Send Data	<i>None</i>
Receive Data	112 bytes data as defined below
Access Level	<i>NONE</i>

Get Cycle Counters Structure			Size: 112 bytes
Type	Field	Description	
0 ULONG	CycleRecordNum	Record Number Default: 0 Displays the highest record number recorded by the system	
4 ULONG	CycleCounts	Total Cycles Default: 0 Displays the total number of operational cycles performed by the system. These cycles could consist of dispense, priming, power-on, self-test or any other operational cycle. This value cannot be reset by the user.	
8 ULONG	PowerCycles	Total Power Cycles Default: 0 Displays the total number of power-on cycles performed by the system. This consists only of power-on cycles. This value cannot be reset by the user.	
12 ULONG	DispenseVolume	Total Dispensed Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume dispensed by the system. This value cannot be reset by the user.	
16 ULONG	FiltratedVolume	Total Filtrated Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume filtrated by the system. This value cannot be reset by the user.	
20 ULONG	VentedVolume	Total Vented Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume vented by the system. This value cannot be reset by the user.	
24 ULONG	PurgedVolume	Total Purged Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume purged by the system. This value cannot be reset by the user.	

28	ULONG	FilledVolume	Total Filled Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume filled into the system. This value cannot be reset by the user.
32	ULONG	CycleCounts_reset_A	Resettable A Cycles Default: 0 Displays the total number of operational cycles performed by the system. These cycles could consist of dispense, priming, power-on, self-test or any other operational cycle. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters A.
36	ULONG	PowerCycles_reset_A	Resettable A Power Cycles Default: 0 Displays the total number of power-on cycles performed by the system. This consists only of power-on cycles. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters A.
40	ULONG	DispenseVolume_reset_A	Resettable A Dispensed Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume dispensed by the system. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters A.
44	ULONG	FiltratedVolume_reset_A	Resettable A Filtrated Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume filtrated by the system. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters A.
48	ULONG	VentedVolume_reset_A	Resettable A Vented Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume vented by the system. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters A.
52	ULONG	PurgedVolume_reset_A	Resettable A Purged Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume purged by the system. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters A.
56	ULONG	FilledVolume_reset_A	Resettable A Filled Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume filled into the system. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters A.

60	ULONG	CycleCounts_reset_B	Resettable B Cycles Default: 0 Displays the total number of operational cycles performed by the system. These cycles could consist of dispense, priming, power-on, self-test or any other operational cycle. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters B.
64	ULONG	PowerCycles_reset_B	Resettable B Power Cycles Default: 0 Displays the total number of power-on cycles performed by the system. This consists only of power-on cycles. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters B.
68	ULONG	DispenseVolume_reset_B	Resettable B Dispensed Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume dispensed by the system. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters B.
72	ULONG	FiltratedVolume_reset_B	Resettable B Filtrated Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume filtrated by the system. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters B.
76	ULONG	VentedVolume_reset_B	Resettable B Vented Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume vented by the system. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters B.
80	ULONG	PurgedVolume_reset_B	Resettable B Purged Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume purged by the system. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters B.
84	ULONG	FilledVolume_reset_B	Resettable B Filled Volume (L) Default: 0 Units: * 1000 Displays the cumulative total of volume filled into the system. This value can be reset by the user by using the RESET_CYCLES command and resetting cycle counters B.
88	ULONG	Reserved1	Reserved Default: 0
92	ULONG	Reserved2	Reserved Default: 0

96	ULONG	Reserved3	Reserved Default: 0
100	ULONG	Reserved4	Reserved Default: 0
104	ULONG	Reserved5	Reserved Default: 0
108	UINT16	Reserved10	For alignment Default: 0
110	UINT16	CRC	CRC Default: 0

VOLUMECOUNTERS: Get Volume Counters Size: 24 bytes

Mini pumps structure with the exactly same command number

Command Code: 50 VOLUMECOUNTERS	
Send Data	<i>None</i>
Receive Data	24 bytes data as defined below
Access Level	<i>NONE</i>

Get Volume Counters Structure			Size: 24 bytes
Type	Field	Description	
0 ULONG	StartCycle	Start Cycle Number Default: 0 What cycle did the counters begin counting (reset)	
4 ULONG	Dispensed	Dispensed Volume (uL) Default: 0 Units: uL uL Volume through nozzle	
8 ULONG	SoftTrigger	MMI Triggered Volume (uL) Default: 0 Units: uL uL Volume from MMI triggers	
12 ULONG	Purged	Purged Volume (uL) Default: 0 Units: uL uL How much purged	
16 ULONG	Vented	Vented Volume (uL) Default: 0 Units: uL uL How much vented	
20 ULONG	Reserved1	Reserved Default: 0	

RESETVOLUMECOUNTERS: Reset Volume Counters

This command is used to reset the volume counters (see VOLUMECOUNTERS).

Command Code: 51 RESETVOLUMECOUNTERS	
Send Data	None
Receive Data	None
Access Level	OPERATOR

SYSTEMNAMES: System Names Size: 288 bytes

System names are strings stored in the pump as a convenience to identify the pump. The strings can be read back with SVIDs as well.

Command Code: 64 SYSTEMNAMES	
Send Data	None
Receive Data	288 bytes data as defined below
Access Level	OWNER

Command Code: 65 WRITE SYSTEMNAMES	
Send Data	288 bytes data as defined below
Receive Data	None
Access Level	OWNER

System Names Structure			Size: 288 bytes
Type	Field	Description	
0 STRING	CustomerName[24]	Customer Name Default: "End user name"	
24 STRING	FabName[24]	Fab Name Default: "Fab name"	
48 STRING	ToolName[24]	Tool Name Default: "Tool name"	
72 STRING	CoaterName[24]	Coater Name Default: "Coater name"	
96 STRING	NozzleName[24]	Nozzle Name Default: "Nozzle name"	
120 STRING	ResistName[24]	Resist Name Default: "Resist name"	
144 STRING	PumpName[24]	Pump Name Default: "Entegris Pump"	
168 STRING	FluidName[24]	Fluid Name/Type Default: "Fluid Name"	
192 STRING	Reserved1[24]	Reserved Default: "Reserved"	
216 STRING	Reserved2[24]	Reserved Default: "Reserved"	
240 STRING	Reserved3[24]	Reserved Default: "Reserved"	
264 STRING	Reserved4[24]	Reserved Default: "Reserved"	

SYSTEM_PARAMETERS: System Parameters Size: 112 bytes

This command reads and writes the System parameters.

Command Code: 66 SYSTEM_PARAMETERS	
Send Data	<i>None</i>
Receive Data	112 bytes data as defined below
Access Level	ENGINEER

Command Code: 67 WRITE SYSTEM_PARAMETERS	
Send Data	112 bytes data as defined below
Receive Data	<i>None</i>
Access Level	ENGINEER

System Parameters Structure		Size: 112 bytes
Type	Field	Description
0 UINT16	Address	Serial Address Min: 1 Max: 63 Default: 1
2 UINT16	FluidViscosity_cP	Fluid Viscosity Min: 1 Max: ST_MAX_VISCOSITY_cP Default: 30 Units: cP This should be set to the viscosity of the fluid in use.
4 UINT16	ReadyPressureCorrectionDisable	RDY Pressure Volume Adj Max: 1 Default: 0 Values: Enable=0, Disable=1 This will disable the dispense volume adjustment that is related to ready pressure correction. If this setting is disabled, the dispense volume will be the exact volume that is entered in the recipe. If this setting is enabled, the dispense volume will be adjusted by the amount of the ready pressure correction.
6 UINT16	ReadyPressureEnable	Ready Pressure Enable Max: 1 Default: 1 Values: Disable=0, Enable=1

8	LONG	ReadyPressureSetpoint_mPSI	RDY Pressure Setpoint Min: ST_READY_PRS_MIN_mPSI Max: 10000 Default: 3000 Units: psi * 1000 This sets the system pressure at ""Ready"".
12	UINT16	reserved2	reserved Default: 0
14	UINT16	OutletValveEnable	Outlet Valve Enable Max: 1 Default: 1 Values: Disable=0, Enable=1 This enables and disables the pneumatics for the external outlet valve.
16	UINT16	ExternalRecValveEnable	External Rec Valve Enable Max: 1 Default: 0 Values: Disable=0, Enable=1 This enables and disables external recirculation valve. If present, the external valve will be used for all recirculation features.
18	UINT16	AutoRecEnable	Auto-Recirculation Max: 1 Default: 0 Values: Disable=0, Enable=1 This enables and disables automatic recirculation of the system defined by the ""Auto-Recirculation Volume"", ""Auto-Recirculation Rate"" and ""Auto-Recirculation Timer"" parameters.
20	UINT16	AutoRecVolume_uL	Auto-Recirc. Volume Min: 500 Max: ST_MAX_DISPENSE_VOLUME Default: 3000 Units: ml * 1000 This sets the volume for the automatic recirculation feature.
22	UINT16	AutoRecRate_uLS	Auto-Recirc. Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the rate for the automatic recirculation feature.
24	LONG	AutoRecTimer_min	Auto-Recirc. Timer Min: 0 Max: 1000 Default: 15 Units: min This sets the frequency (in minutes) automatic recirculation occurs.

28	UINT16	InCycleAutoRecRate_uLS	In-cycle Auto-Recirc Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the rate for the full ""auto-recirculation"" within every dispense cycle.
30	UINT16	InCycleAutoRec	In-cycle Auto-Recirc Enable Max: 1 Default: 0 Values: Disable=0, Enable=1 This enables and disables full ""auto-recirculation"" within every dispense cycle. This is separate from the standard ""Auto-Recirculation"" function (which is meant to run during idle periods) and can be used independently. This will purge the full remaining fluid in the dispense chamber before filtration.
32	UINT16	MaintenanceModeTimeoutEnable	Maint. Mode Timeout Max: 1 Default: 0 Values: Disable=0, Enable=1 This enables and disables the maintenance mode feature. When enabled, communication with the unit is maintained but other operations are disabled for the amount of time defined by the ""Maintenance Mode Timeout"" in minutes.
34	UINT16	MaintenanceModeTimeout_min	Maint. Mode Timeout Max: 1000 Default: 120 Units: min This sets the length of time (in minutes) the maintenance mode is enabled.
36	LONG	reserved5	reserved Default: 0
40	ULONG	FilterCounterLimit	Filter Cycle Alarm Limit Max: 10000000 Default: 250000 Units: cycles This sets a cycles limit for the filter which is tracked by ""Resettable Cycle Counter A"". An alarm will be generated when this value is exceeded. Set the limit to ""0"" to disable this feature

44	UINT16	FilterCounterAlarmLevel	Filter Cycle Alarm Level Min: 1 Max: 3 Default: 2 Values: Information=1, Warning=2, Error=3 This selects the severity level for filter cycle counter alarms. If the cycle counter on the filter exceeds the limit set by the user, these alarms will be posted. ""Information"", will post an event only. ""Warning"", will post an alarm but allow normal operation to continue. ""Error"", will finish the current cycle then hold the system in an error state until cleared. ""Critical Error"" will instantly put the pump in error state which will require an initialization.
46	UINT16	ConsecutiveAutoVentLimit	Consecutive Auto-Vent Limit Max: 1000 Default: 100 Units: cycles This sets a limit for the number of consecutive auto vents performed by the system. This alerts the user to a system that is performing a large number of unintended auto-vents. An alarm can be generated if this limit is exceeded.
48	UINT16	ConsecutiveAutoVentAlarmLevel	Consecutive Auto-Vent Alarm Max: 3 Default: 2 Values: none=0, Information=1, Warning=2, Error=3 This selects the severity level for alarms generated by consecutive auto vents. If the number of consecutive auto vents exceeds the limit set by the user, these alarms will be posted. ""Information"", will post an event only. ""Warning"", will post an alarm but allow normal operation to continue. ""Error"", will finish the current cycle then hold the system in an error state until cleared. ""Critical Error"" will instantly put the pump in error state which will require an initialization.
50	UINT16	reserved7	reserved Default: 0
52	UINT16	SerialTriggerEnable	Serial Trigger Enable Max: 1 Default: 1 Values: Disable=0, Enable=1 This enables and disables the serial command trigger. When enabled, dispense recipes and priming recipes can be triggered from the MMI software.
54	UINT16	reserved8	reserved Default: 0

56	UINT16	VentOnlyVolume_uL	Vent Only Volume Min: 100 Max: ST_MAX_DISPENSE_VOLUME Default: 500 Units: ml * 1000 This sets the volume for the ""Vent Only"" feature.
58	UINT16	VentOnlyRate_uLS	Vent Only Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the rate for the ""Vent Only"" feature.
60	LONG	reserved9	reserved Default: 0
64	UINT16	PreFiltrationAlarmLevel	Pre-Filtration Alarm Level Min: 1 Max: 3 Default: 2 Values: Information=1, Warning=2, Error=3 This selects the severity level for the prefiltration alarm. ""Warning"", will post an alarm but allow normal operation to continue. ""Error"", will finish the current cycle then hold the system in an error state until cleared.
66	UINT16	FiltrationAlarmLevel	Filtration Alarm Level Min: 2 Max: 3 Default: 2 Values: Warning=2, Error=3 This selects the severity level for any filtration alarms. ""Warning"", will post an alarm but allow normal operation to continue. ""Error"", will finish the current cycle then hold the system in an error state until cleared.
68	UINT16	UtilityAlarmLevel	Utility Alarm Level Min: 1 Max: 3 Default: 2 Values: Information=1, Warning=2, Error=3 This selects the severity level for utility alarms. If the N2 supply or vacuum utility drop below spec, these alarms will be posted. ""Information"", will post an event only. ""Warning"", will post an alarm but allow normal operation to continue. ""Error"", will finish the current cycle then hold the system in an error state until cleared.

70	UINT16	ReadyPressureAlarmLevel	Ready Pressure Alarm Level Min: 2 Max: 3 Default: 2 Values: Warning=2, Error=3 This selects the severity level for ready pressure alarms. If the ready pressure drop out of spec for more then 1 second, these alarms will be posted. ""Warning"", will post an alarm but allow normal operation to continue. ""Error"", will finish the current cycle then hold the system in an error state until cleared.
72	INT16	PE_DisDelay_mS	Partial Emulation Dis Delay Min: 0 Max: 500 Default: 0 Units: ms This sets the dispense delay (in seconds) for partial emulation interface models.
74	UINT16	HPT_StepSize	HPT Pump Size Max: 1 Default: 0 Values: 10cc_version=0, 6cc_version=1 Used for HPT retrofit, this value selects the size of the HPT pump that was replaced. For all other installations this field has no effect.
76	UINT16	FlowSensorEnable	Flow Sensor Enable Max: 1 Default: 0 Values: Disable=0, Enable=1 This enables and disables the flow sensor, if present. For other units, this field has no effect.
78	UINT16	Suckback_mode	Suckback Mode Max: 1 Default: 0 Values: Add_Volume=0, Don't_Add=1 This sets the suckback mode for all dispense recipes. In mode ""Add Volume"", the suckback volume will be added to the dispense volume. This is typical and backward compatible with older firmware versions. In mode ""Don't Add"", the suckback volume will NOT be added to the dispense volume. In this mode the Suckback volume must be lower than the dispense volume. Otherwise the system will error at the time of dispense.

80	UINT16	PreDispense_mode	Pre-Dispense Mode Max: 1 Default: 0 Values: Separate=0, Timed Pre-Dispense=1 This sets the pre-dispense mode for all dispense recipes. In mode ""Timed Pre-Dispense"", A pre-dispense will be executed on the first trigger if less than .7 seconds, after which the second trigger will execute a main dispense. For first triggers greater than .7 seconds a main will execute with no pre-dispense. In mode ""Separate"", The system will operate normally.
82	UINT16	FlowCalibration	Flow Calibration Min: 100 Max: 1900 Default: 1000 Units: % * 10 The Total Flow Volume is adjusted by this percent. 100% is no change.
84	UINT16	PreDispense_timeout	Pre-Dispense Timeout Min: 15 Max: 300 Default: 30 Units: s Maximum time allowed between pre-dispense and main dispense before alarming (applies to HV20 pump only, all others are fixed at 30 seconds)
86	UINT16	LowerMaxDispPressure	Lower maximum dispense pressure flag Max: 1 Default: 0
88	INT16	FiltrationSetup_PSI_Control_Enable	Filtration Setup Ctrl Enable Min: 0 Max: 1 Default: 0 Values: Disable=0, Enable=1 This enables or disables the system to control fill chamber pressure starting at the ready state until start of filtration.
90	UINT16	DCMissingRefAlarmLevel	DC: No Reference Alarm Level Min: 1 Max: 3 Default: 2 Values: Information=1, Warning=2, Error=3 This selects the severity level of the alarm that is posted when there is no Dispense Confirmation reference cycle saved. ""Information"", will post an event only. ""Warning"", will post an alarm but allow normal operation to continue. ""Error"", will finish the current cycle then hold the system in an error state until cleared.

92 LONG	FiltrationSetupPSISetpoint_mPSI	Filtration Setup Ctrl Setpoint Min: 2000 Max: 10000 Default: 4000 Units: psi * 1000 This sets the fill chamber control pressure starting at the ready state until start of filtration.
96 LONG	reserved24	reserved Default: 0
100 LONG	reserved25	reserved Default: 0
104 ULONG	Time	Last saved Default: 0 Displays the date and time when this structure was last saved
108 ULONG	Cycle	Cycle when written Default: 0

RECHARGE: Recharge Parameters Size: 104 bytes

These commands are used to read and write the stored recharge parameters. These values are used in the process of recharging and refilling the system.

Command Code: 70 RECHARGE	
Send Data	None
Receive Data	104 bytes data as defined below
Access Level	ENGINEER

Command Code: 71 WRITE RECHARGE	
Send Data	104 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Recharge Parameters Structure		Size: 104 bytes
Type	Field	Description
0 UINT16	reserved1	reserved Default: 0
2 INT16	AutoVentEnable	Auto-Vent Enable Min: 0 Max: 1 Default: 1 Values: Disable=0, Enable=1 This enables and disables automatic venting
4 INT16	VentFreqEnable	Vent Frequency Enable Min: 0 Max: 1 Default: 1 Values: Disable=0, Enable=1 This enables and disables vent frequency defined by the ""Vent Frequency"", ""Vent Volume"" and ""Vent Rate"" parameters below.
6 UINT16	VentFreq	Vent Frequency Min: 1 Max: 10000 Default: 1 Units: cycles This sets the frequency (in cycles) at which venting occurs.

8	UINT16	FreqVentVolume	Vent Volume (Frequency) Min: 100 Max: 3000 Default: 500 Units: ml * 1000 This sets the volume for the vent segment. This used for venting that was caused by vent frequency settings.
10	UINT16	FreqVentRate	Vent Rate (Frequency) Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the rate for the vent segment. This used for venting that was caused by vent frequency settings.
12	UINT16	AutoVentVolume	Vent Volume (Auto-Vent) Min: 100 Max: 3000 Default: 500 Units: ml * 1000 This sets the volume for the vent segment. This used for venting that was caused by auto-vent settings.
14	UINT16	AutoVentRate	Vent Rate (Auto-Vent) Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the rate for the vent segment. This used for venting that was caused by auto-vent settings.
16	UINT16	PurgeVolume_uL	Purge Volume Min: 300 Max: ST_MAX_PURGE_VOLUME Default: 400 Units: ml * 1000 This sets the volume for the purge segment.
18	UINT16	PurgeRate	Purge Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the rate for the purge segment.
20	LONG	reserved3	reserved Default: 0

24	INT16	FillAsRequired	Fill As Required Min: 0 Max: 1 Default: 0 Values: Every Cycle=0, As Required=1 This selects when the pump will fill. Fill completely on every cycle or fill only when required based on ""Max. Dispense Volume"".
26	INT16	AirTestEnable	Air Detect Enable Min: 0 Max: 1 Default: 1 Values: Disable=0, Enable=1 This enables and disables automatic air detection testing during recharge. (If set to ""Disable"", a result of ""9.99 mL"" air detection will be reported to confirm test is set to Disable and not protecting the system.)
28	INT16	VarFillCurrent	Fill Variable Current Min: 0 Max: 1 Default: 0 Values: Disable=0, Enable=1 This enables and disables control of the system's recharge by lowering the fill current based determined by pressure load.
30	INT16	FillPressureEnable	Fill Pressure Cntrl Enable Min: 0 Max: 1 Default: 1 Values: Disable=0, Enable=1 This enables and disables control of the system's recharge through the ""Fill Pressure Minimum"". When disabled, the fill motor drives at a constant rate during the fill segment.
32	INT16	Fill_VariableVolume	Fill Alternate Home Min: 0 Max: 1 Default: 0 Values: Disable=0, Enable=1 This enables and disables the system to use the variable hold-up volume during fill. The variable hold-up volume is determined by the ""Max. Dispense Volume"".
34	UINT16	FillRate	Fill Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the rate for the fill segment.
36	LONG	reserved6	reserved Default: 0

40 LONG	FillPressureMin_mPSI	Fill Pressure Minimum Min: -12000 Max: -1000 Default: "-10000" Units: psi * 1000 This sets the minimum fill pressure for the fill segment. The fill will happen at the programmed fill rate until this pressure limit is exceeded at which point the fill motor will slow down so this pressure limit will be exceeded.
44 LONG	reserved7	reserved Default: 0
48 UINT16	FiltrationRate	Filtration Rate Min: 10 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the rate for the filtration recharge segment.
50 UINT16	reserved8	reserved Default: 0
52 LONG	FiltPressureMax_mPSI	Filtration Pressure Max Min: 5000 Max: ST_MAX_FILT_PRESSURE_mPSI Default: 30000 Units: psi * 1000 This sets the maximum upstream pressure during the filtration segment. If maximum pressure exceeds this value, an alarm will be posted. Reference the ""Filtration Alarm Level"" on the Systems page to set alarm type.
56 LONG	FiltrationPressureSetpoint	Filtration psi Setpoint Min: 2000 Max: 6000 Default: 4000 Units: psi * 1000 This sets the pressure during the filtration recharge segment.
60 UINT16	reserved9	reserved Default: 0
62 UINT16	VariableDispenseVolume	Max. Dispense Volume Min: 1000 Max: ST_MAX_DISPENSE_VOLUME Default: 3000 Units: ml * 1000 This sets the system's maximum dispense volume.

64 INT16	ReadyFillDisable	Ready Fill Enable Min: 0 Max: 1 Default: 0 Values: Enable=0, Disable=1 This enables or disables the system to fill during the ready state. When disabled, the fill motor is require to finish the filling state before the system goes ready. When disable, the system can finish the filling during the ready state.
66 UINT16	reserved20	reserved Default: 0
68 LONG	reserved21	reserved Default: 0
72 LONG	reserved22	reserved Default: 0
76 LONG	reserved23	reserved Default: 0
80 LONG	reserved24	reserved Default: 0
84 LONG	reserved25	reserved Default: 0
88 LONG	reserved26	reserved Default: 0
92 LONG	reserved27	reserved Default: 0
96 ULONG	Time	Last saved Default: 0 Displays the date and time when this structure was last saved
100 ULONG	Cycle	Cycle when written Default: 0

START_RECHARGE: Start a Recharge

This command will start a recharge of the pump.

Command Code: 73 START_RECHARGE	
Send Data	None
Receive Data	None
Access Level	OPERATOR

START_RECIPE: Recipe Start Size: 4 bytes

This comand will start the recipe number that is sent in the 2 byte value sent with the command.

Command Code: 72 START_RECIPE	
Send Data	2 bytes for Recipe Number
Receive Data	None
Access Level	OPERATOR

Recipe Start Structure			Size: 4 bytes
Type	Field	Description	
0 UINT16	RecipeNum	Start Recipe Min: 1 Max: 12 Default: 1	
2 UINT16	reserved1	reserved1 Default: 0	

RECIPE: Recipe Size: 124 bytes

This command will get and write any of the stored recipes.

Command Code: 74 RECIPE	
Send Data	2 bytes for Recipe Number
Receive Data	124 bytes data as defined below
Access Level	ENGINEER

Command Code: 75 WRITE RECIPE	
Send Data	124 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Recipe Structure		Size: 124 bytes
Type	Field	Description
0 UINT16	RecipeNum	Recipe number Max: 12 Default: 1 This selects the recipe number.
2 UINT16	reserved0	reserved0 Default: 0
4 STRING	RecipeTitle[24]	Description Default: "Recipe Title" This sets the name for the corresponding recipe number.
28 INT16	OpenAdjust	Outlet Valve Open Adj. Min: -3000 Max: 3000 Default: 0 Units: ms This sets the adjustment time to open the outlet valve prior to dispense motor movement.
30 UINT16	DisRampUpTime	Ramp Up Time Max: 500 Default: 0 Units: ms This sets motor ramp up adjustment time from zero to the dispense rate.
32 ULONG	reserved1	reserved Default: 0
36 ULONG	reserved2	reserved Default: 0

40	UINT16	DispenseVolume_1	Segment 1 Volume Max: ST_MAX_DISPENSE_VOLUME Default: 3000 Units: ml * 1000 This sets the volume for dispense segment 1.
42	UINT16	DispenseRate_1	Segment 1 Rate Max: 3000 Default: 1000 Units: ml/s * 1000 This sets the rate for dispense segment 1.
44	UINT16	DispenseVolume_2	Segment 2 Volume Max: ST_MAX_DISPENSE_VOLUME Default: 0 Units: ml * 1000 This sets the volume for dispense segment 2.
46	UINT16	DispenseRate_2	Segment 2 Rate Max: 3000 Default: 0 Units: ml/s * 1000 This sets the rate for dispense segment 2.
48	UINT16	DispenseVolume_3	Segment 3 Volume Max: ST_MAX_DISPENSE_VOLUME Default: 0 Units: ml * 1000 This sets the volume for dispense segment 3.
50	UINT16	DispenseRate_3	Segment 3 Rate Max: 3000 Default: 0 Units: ml/s * 1000 This sets the rate for dispense segment 3.
52	UINT16	DispenseVolume_4	Segment 4 Volume Max: ST_MAX_DISPENSE_VOLUME Default: 0 Units: ml * 1000 This sets the volume for dispense segment 4.
54	UINT16	DispenseRate_4	Segment 4 Rate Max: 3000 Default: 0 Units: ml/s * 1000 This sets the rate for dispense segment 4.
56	UINT16	DispenseVolume_5	Segment 5 Volume Max: ST_MAX_DISPENSE_VOLUME Default: 0 Units: ml * 1000 This sets the volume for dispense segment 5.

58	UINT16	DispenseRate_5	Segment 5 Rate Max: 3000 Default: 0 Units: ml/s * 1000 This sets the rate for dispense segment 5.
60	ULONG	reserved3	reserved Default: 0
64	ULONG	reserved4	reserved Default: 0
68	UINT16	DisRampDownTime	Ramp Down Time Max: 500 Default: 0 Units: ms This sets motor ramp down adjustment time from the dispense rate to zero.
70	INT16	CloseAdjust	Outlet Valve Close Adj. Min: -3000 Max: 3000 Default: 0 Units: ms This sets the adjustment time to close the outlet valve after dispense motor movement.
72	UINT16	SuckBackDelay_mS	Suck-back Delay Max: 10000 Default: 0 Units: ms This sets the suckback delay time in milliseconds
74	UINT16	SuckBackVolume_uL	Suck-back Volume Max: ST_MAX_SUCKBACK_VOLUME Default: 0 Units: ml * 1000 This sets the volume for the suckback segment.
76	UINT16	SuckBackRate_uLs	Suck-back Rate Max: 3000 Default: 0 Units: ml/s * 1000 This sets the rate for the suckback segment.
78	UINT16	reserved5	reserved Default: 0
80	UINT16	PreDispenseRecipe	Pre-Dispense Recipe Max: 12 Default: 0 This assigns a recipe to be used as pre-dispense.

82 INT16	VolumeCalibration	Volume Calibration Min: ST_MIN_VOLUME_CALIBRATION Max: ST_MAX_VOLUME_CALIBRATION Default: 0 Units: ml * 1000 This sets a volume to be added or subtracted from the total dispense volume.
84 ULONG	DisplayTime_ms	Calculated total dispense time Default: 0 Units: s * 1000 Displays the calculated total dispense time for the selected recipe.
88 ULONG	DisplayVoume_uL	Calculated total dispense time Default: 0 Units: ml * 1000 Displays the calculated total dispense volume for the selected recipe.
92 LONG	reserved22	reserved Default: 0
96 LONG	reserved23	reserved Default: 0
100 LONG	reserved24	reserved Default: 0
104 LONG	reserved25	reserved Default: 0
108 LONG	reserved26	reserved Default: 0
112 LONG	reserved27	reserved Default: 0
116 ULONG	Time	Last saved Default: 0 Displays the date and time when this structure was last saved
120 ULONG	Cycle	Cycle when written Default: 0

START_PRIMING: Priming Start Size: 4 bytes

Start the priming sequence based on the number being set to the system. See the [Appendix](#) for more detailed information about this command.

Command Code: 76 START_PRIMING	
Send Data	2 bytes Sequence Number
Receive Data	None
Access Level	ENGINEER

Priming Start Structure			Size: 4 bytes
Type	Field	Description	
0 INT16	SequenceNum	SequenceNum Min: -2 Max: 20 Default: 1	
2 INT16	reserved1	reserved1 Default: 0	

PRIMING: Priming Sequence Size: 864 bytes

This command is used to read and write the different priming sequences stored in the system. See the [Appendix](#) for more detailed information about this command.

Command Code: 78 PRIMING	
Send Data	2 bytes Sequence Number
Receive Data	864 bytes data as defined below
Access Level	ENGINEER

Command Code: 79 WRITE PRIMING	
Send Data	864 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Priming Sequence Structure			Size: 864 bytes
Type	Field	Description	
0 UINT16	SequenceNum	Sequence Number Max: 20 Default: 1 Units: SeqNum	
2 UINT16	reserved0	reserved Default: 0	
4 STRING	SequenceTitle[32]	Sequence Title Default: "Sequence Title" This sets the name for the corresponding priming sequence.	
36 STRING	SequenceDescription[80]	Sequence Description Default: "User Sequence Description" This priming sequence is used best for [fluid] or [filter].	
116 UINT16	StepType[30]	Cycle Type This lists the [9] priming functions . Max: 11 Default: 0 Units: type	
176 UINT16	StepCycleCount[30]	Number Cycles Min: 1 Max: 10000 Default: 1 Units: counts This sets the cycle number of each cycle type.	

236	UINT16	VentRate[30]	Vent Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the vent rate during this step.
296	UINT16	OutletRate[30]	Outlet Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the outlet rate during this step.
356	UINT16	PurgeRate[30]	Purge Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the vent rate during this step.
416	UINT16	FillRate[30]	Fill Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the purge rate during this step.
476	UINT16	FiltrationRate[30]	Filtration Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the filtration rate during this step.
536	UINT16	GotoStep[30]	Loop Min: 1 Max: 30 Default: 1 Units: step This ""go to"" function sets the sequence back to a specified.
596	INT16	FiltrationPressure_cPSI[30]	Filtration Pressure cPSI Min: 50 Max: 2000 Default: 300 Units: psi * 100 This sets the filtration pressure during this step.
656	INT16	SoakPressure_cPSI[30]	Filter Soak Pressure cPSI Min: -1000 Max: ST_SOAK_PRESSURE_MAX Default: 1000 Units: psi * 100 This sets the soak pressure during this backflush step.

716	UINT16	SoakTime_min[30]	Filter Soak Time Min: 1 Max: 120 Default: 5 Units: min This sets the filter soak time during this backflush step.
776	UINT16	reserved[30]	reserved Default: 0
836	ULONG	Time	Date and time when written Default: 0
840	ULONG	Cycle	Cycle when written Default: 0
844	ULONG	EstTime_Sec	Est Total time to complete Default: 0 Units: s Displays the estimated time for this priming sequence
848	ULONG	EstOutletVol_uL	Est total volume out the outlet Default: 0 Units: ml * 1000 Displays the estimated volume going to the outlet during this priming sequence
852	ULONG	EstVentVol_uL	Est total volume out the vent Default: 0 Units: ml * 1000 Displays the estimated volume going to the vent during this priming sequence
856	ULONG	EstFilteredVol_uL	Est total volume through the filter Default: 0 Units: ml * 1000 Displays the estimated volume going through the filter during this priming sequence
860	ULONG	reserved2	reserved Default: 0

PRIMINGSTATUS: Priming Status Size: 52 bytes

This command returns the status of an active priming sequence. If the Active field is 0, no priming is active. If the Active field is 1, then a priming sequence is active and the rest of the information is valid. See the [Appendix](#) for more detailed information about this command.

Command Code: 80 PRIMINGSTATUS	
Send Data	None
Receive Data	52 bytes data as defined below
Access Level	NONE

Priming Status Structure			Size: 52 bytes
Type	Field	Description	
0 UINT16	SequenceNum	Which Sequence is running Default: 0	
2 UINT16	Active	Is Priming Active Default: 0	
4 ULONG	StartTime	time of priming start Default: 0 Units: s	
8 UINT16	RealStepNumber	Whats the real step including loops Default: 0	
10 UINT16	RealStepType	What type of step is the the real step Default: 0	
12 UINT16	RealStepTotalCycles	Total cycles in the real step Default: 0	
14 UINT16	RealStepCyclesRemaining	cycles remaining in the real step Default: 0	
16 UINT16	RuningStepNumber	What step is running now Default: 0	
18 UINT16	RuningStepType	What type of step is running now Default: 0	
20 UINT16	RuningStepTotalCycles	Total cycles in the step running now Default: 0	
22 UINT16	RuningStepCyclesRemaining	cycles remaining in the step running now Default: 0	
24 ULONG	TotalTime_sec	Total priming time Default: 0 Units: s	
28 ULONG	RemianingTime_sec	Remaining priming time Default: 0 Units: s	

32	UINT16	PercentComplete	Percent Complete Default: 0 Units: %
34	UINT16	Paused	Is the Priming paused Default: 0
36	ULONG	reserved3	reserved Default: 0 Units: s
40	ULONG	reserved4	reserved Default: 0
44	ULONG	reserved5	reserved Default: 0 Units: s
48	ULONG	reserved6	reserved Default: 0

PRIMINGSTATUS_SEMES: Priming Status Semes Size: 28 bytes

This command returns the status of an active priming sequence. If the Active field is 0, no priming is active. If the Active field is 1, then a priming sequence is active and the rest of the information is valid. See the [Appendix](#) for more detailed information about this command.

Command Code: 81 PRIMINGSTATUS_SEMES	
Send Data	None
Receive Data	28 bytes data as defined below
Access Level	NONE

Priming Status Semes Structure			Size: 28 bytes
Type	Field	Description	
0 UINT16	SequenceNum	Which Sequence is running Default: 0	
2 UINT16	RealStepNumber	Whats the real step including loops Default: 0	
4 UINT16	RealStepType	What type of step is the the real step Default: 0	
6 UINT16	RealStepTotalCycles	Total cycles in the real step Default: 0	
8 UINT16	RealStepCyclesRemaining	Cycles remaining in the real step Default: 0	
10 UINT16	RuningStepNumber	What step is running now Default: 0	
12 UINT16	RuningStepType	What type of step is running now Default: 0	
14 UINT16	RuningStepTotalCycles	Total cycles in the step running now Default: 0	
16 UINT16	RuningStepCyclesRemaining	Cycles remaining in the step running now Default: 0	
18 UINT16	PercentComplete	Percent Complete Default: 0 Units: %	
20 ULONG	TotalTime_sec	Total priming time Default: 0 Units: s	
24 ULONG	RemianingTime_sec	Remaining priming time Default: 0 Units: s	

RTSTATUS: Get Real Time Status Size: 76 bytes

Command Code: 100 RTSTATUS	
Send Data	None
Receive Data	76 bytes data as defined below
Access Level	NONE

Get Real Time Status Structure		Size: 76 bytes
Type	Field	Description
0 INT16	Status	System Status Default: 0 Test
2 UINT16	Indicators	Status bit indicators Default: 0
4 ULONG	LastDispenseCycle	Last Dispensed Cycle Default: 0
8 UINT16	LastRecipeNum	Last Dispensed Recipe Num Default: 0
10 UINT16	LastDispenseVol_uL	Last Dispensed Total volume Default: 0
12 UINT16	LastDispenseConfirmation_per	Last Dispensed confirmation value Default: 0
14 UINT16	reserved1	reserved Default: 0
16 ULONG	TotalCycleCounts	Cycle Default: 0
20 ULONG	ResettableCycleCounts	Cycle on Filter Default: 0
24 ULONG	ResettableFiltratedVolume	Volume on Filter Default: 0 Units: L * 1000
28 UINT16	CriticalErrors	counts Critical System Errors after they are logged Default: 0
30 UINT16	Errors	counts System Errors after they are logged Default: 0
32 UINT16	Warnings	counts System Warnings after they are logged Default: 0
34 UINT16	Infos	counts System infos after they are logged Default: 0
36 UINT16	TotalAlarms	counts total uncleared alarms after they are logged Default: 0

38	UINT16	reserved2	reserved Default: 0
40	ULONG	NewestAlarmNum	Number of newest alarm Default: 0
44	ULONG	NewestRecordNum	Number of newest record Default: 0
48	ULONG	LastClearedRecord	# of the last cleared record Default: 0
52	ULONG	LastClearedAlarm	# of the last cleared alarm Default: 0
56	ULONG	ProfileCycle	Most recent profile cycle Default: 0
60	INT16	N2Sensor_cPSI	N2 Sensor Default: 0 Units: psi * 100
62	INT16	VacSensor_cPSI	Vacuum Sensor Default: 0 Units: psi * 100
64	UINT16	OfflineState	OfflineState Default: 0 The system online/offline state
66	UINT16	ExternalRecValveEnable	External Rec Valve Enable Default: 0
68	LONG	ConsecutiveAutoVents	ConsecutiveAutoVents Default: 0 This is the number of consecutive auto vents performed by the system. This alerts the user to a system that is performing a large number or unintended auto vents.
72	LONG	reserved5	reserved Default: 0

CS_RTSTATUS: Get Real Time Status for Cyberspace Size: 124 bytes

Command Code: 110 CS_RTSTATUS	
Send Data	None
Receive Data	124 bytes data as defined below
Access Level	NONE

Get Real Time Status for Cyberspace StructSize: 124 bytes			
Type	Field	Description	
0 INT16	Status	System Status Default: 0 Test	
2 UINT16	Indicators	Status bit indicators Default: 0	
4 ULONG	LastDispenseCycle	Last Dispensed Cycle Default: 0	
8 UINT16	LastRecipeNum	Last Dispensed Recipe Num Default: 0	
10 UINT16	LastDispenseVol_uL	Last Dispensed Total volume Default: 0	
12 UINT16	LastDispenseConfirmation_per	Last Dispensed confirmation value Default: 0	
14 UINT16	reserved1	reserved Default: 0	
16 ULONG	TotalCycleCounts	Cycle Default: 0	
20 ULONG	ResettableCycleCounts	Cycle on Filter Default: 0	
24 ULONG	ResettableFiltratedVolume	Volume on Filter Default: 0 Units: L * 1000	
28 UINT16	CriticalErrors	counts Critical System Errors after they are logged Default: 0	
30 UINT16	Errors	counts System Errors after they are logged Default: 0	
32 UINT16	Warnings	counts System Warnings after they are logged Default: 0	
34 UINT16	Infos	counts System infos after they are logged Default: 0	
36 UINT16	TotalAlarms	counts total uncleared alarms after they are logged Default: 0	

38	UINT16	reserved2	reserved Default: 0
40	ULONG	NewestAlarmNum	Number of newest alarm Default: 0
44	ULONG	NewestRecordNum	Number of newest record Default: 0
48	ULONG	LastClearedRecord	# of the last cleared record Default: 0
52	ULONG	LastClearedAlarm	# of the last cleared alarm Default: 0
56	ULONG	ProfileCycle	Most recent profile cycle Default: 0
60	INT16	N2Sensor_cPSI	N2 Sensor Default: 0 Units: psi * 100
62	INT16	VacSensor_cPSI	Vacuum Sensor Default: 0 Units: psi * 100
64	UINT16	OfflineState	OfflineState Default: 0 The system online/offline state
66	STRING	Model[24]	Model Name Default: 0
90	STRING	PumpSerialNumber[24]	Pump Serial Number Default: "(None)"
114	UINT16	reserved3	reserved Default: 0
116	LONG	reserved4	reserved Default: 0
120	LONG	reserved5	reserved Default: 0

PRESSURE_STATUS: PressureStatus Size: 68 bytes

Command Code: 104 PRESSURE_STATUS	
Send Data	None
Receive Data	68 bytes data as defined below
Access Level	NONE

PressureStatus Structure		Size: 68 bytes
Type	Field	Description
0 INT16	reserved1	reserved1 Default: 0
2 UINT16	Dis_UnCal_ADC	dispense un-calibrated ADC counts Default: 0
4 FLOAT	Dis_UnCal_PSI	dispense un-calibrated PSI Default: 0
8 LONG	Dis_UnCal_mPSI	dispense un-calibrated mPSI Default: 0
12 LONG	Dis_pressure_Max_Range_mPSI	dispense max range mPSI Default: 0
16 LONG	Dis_pressure_Min_Range_mPSI	dispense min range mPSI Default: 0
20 FLOAT	Dis_pressure_PSI	dispense calibrated PSI Default: 0
24 LONG	Dis_pressure_mPSI	dispense calibrated mPSI Default: 0
28 INT16	Dis_pressure_cPSI	dispense calibrated cPSI Default: 0
30 UINT16	Fill_UnCal_ADC	Fill un-calibrated ADC counts Default: 0
32 FLOAT	Fill_UnCal_PSI	Fill un-calibrated PSI Default: 0
36 LONG	Fill_UnCal_mPSI	Fill un-calibrated mPSI Default: 0
40 LONG	Fill_pressure_Max_Range_mPSI	Fill max range mPSI Default: 0
44 LONG	Fill_pressure_Min_Range_mPSI	Fill min range mPSI Default: 0
48 FLOAT	Fill_pressure_PSI	Fill calibrated PSI Default: 0
52 LONG	Fill_pressure_mPSI	Fill calibrated mPSI Default: 0

56	INT16	Fill_pressure_cPSI	Fill calibrated cPSI Default: 0
58	INT16	reserved2	reserved2 Default: 0
60	ULONG	disPressureSensor_ERRORS	disPressureSensor_ERRORS Default: 0
64	ULONG	fillPressureSensor_ERRORS	fillPressureSensor_ERRORS Default: 0

UTILITY_STATUS: UtilityStatus Size: 44 bytes

Command Code: 106 UTILITY_STATUS	
Send Data	None
Receive Data	44 bytes data as defined below
Access Level	NONE

UtilityStatus Structure		Size: 44 bytes
Type	Field	Description
0 INT16	TempFillSensor_cC	Fill Temperature Default: 0 Units: C * 100
2 INT16	TempDisSensor_cC	Dispense Temperature Default: 0 Units: C * 100
4 INT16	N2SensorTemp_cC	N2 Sensor Temp Default: 0 Units: C * 100
6 INT16	VacSensorTemp_cC	Vacuum Sensor Temp Default: 0 Units: C * 100
8 INT16	N2Sensor_cPSI	N2 Sensor Default: 0 Units: psi * 100
10 INT16	VacSensor_cPSI	Vacuum Sensor Default: 0 Units: psi * 100
12 INT16	HomingCurrent_mA	Peak homing current of DC motor Default: 0 Units: mA
14 INT16	reserved1	reserved1 Default: 0
16 INT16	DisCurrent_UnCal_mA	Dispense Motor Current Default: 0 Units: mA
18 INT16	DispenseCurrent_mA	Dispense Motor Current Default: 0 Units: mA
20 UINT16	N2SensorTemp_raw	N2 Sensor Temp (raw value) Default: 0 Units: C * 100

22	UINT16	VacSensorTemp_raw	Vacuum Sensor Temp (raw value) Default: 0 Units: C * 100
24	UINT16	N2Sensor_raw	N2 Sensor (raw value) Default: 0 Units: psi * 100
26	UINT16	VacSensor_raw	Vacuum Sensor (raw value) Default: 0 Units: psi * 100
28	ULONG	N2Sensor_Serial	N2 Sensor (serial, HYB only) Default: 0
32	ULONG	VacSensor_Serial	Vacuum Sensor (serial, HYB only) Default: 0
36	UINT16	SensorDetect1	Sensor detection mechanism 1 Default: 0
38	UINT16	SensorDetect2	Sensor detection mechanism 2 Default: 0
40	UINT16	SensorDetect3	Sensor detection mechanism 3 Default: 0
42	UINT16	reserved2	reserved2 Default: 0

DC_LIMITS: Dispense Confirmation Limits Size: 252 bytes

Dispense confirmation limits for each reference recipe.

Command Code: 153 DC_LIMITS	
Send Data	2 bytes for Recipe Number
Receive Data	252 bytes data as defined below
Access Level	ENGINEER

Command Code: 154 WRITE DC_LIMITS	
Send Data	252 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Dispense Confirmation Limits Structure			Size: 252 bytes
Type	Field	Description	
0 UINT16	RecipeNum	Recipe number Min: 1 Max: 12 Default: 1 This selects the DC limits for the recipe.	
2 UINT16	Enable	Enable Dispense Limit Checking Max: 1 Default: 0 Values: Disable=0, Enable=1 Global Enable/disable for all confirmation alarms.	
4 INT16	ReadyPressure_WarnLimit	ReadyPressure_WarnLimit Min: 30 Max: 1000 Default: 50 Units: psi * 100 Limit in PSI before warning is triggered	
6 INT16	ReadyPressure_ErrorLimit	ReadyPressure_ErrorLimit Min: 30 Max: 1000 Default: 100 Units: psi * 100 Limit in PSI before error is triggered	

8	UINT16	ReadyPressure_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Ready Pressure test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
10	UINT16	ReadyPressure_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
12	UINT16	Confirmation_WarnLimit	Confirmation_WarnLimit Min: 10 Max: 99 Default: 85 Units: % Limit in percent before warning is triggered
14	UINT16	Confirmation_ErrorLimit	Confirmation_ErrorLimit Min: 10 Max: 99 Default: 80 Units: % Limit in percent before error is triggered
16	UINT16	Confirmation_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Pressure Profile Compare test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
18	UINT16	Confirmation_TestType	test type greater then or less then Min: 1 Max: 1 Default: 1 Test if the value is LESS than the alarm limits
20	UINT16	FlowConfirmation_WarnLimit	FlowConfirmation_WarnLimit Min: 10 Max: 99 Default: 85 Units: % Limit in percent before warning is triggered

22	UINT16	FlowConfirmation_ErrorLimit	FlowConfirmation_ErrorLimit Min: 10 Max: 99 Default: 80 Units: % Limit in percent before error is triggered
24	UINT16	FlowConfirmation_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Pressure Profile Compare test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
26	UINT16	FlowConfirmation_TestType	test type greater then or less then Min: 1 Max: 1 Default: 1 Test if the value is LESS than the alarm limits
28	INT16	PeakPressure_WarnLimit	PeakPressure_WarnLimit Min: 50 Max: 1000 Default: 100 Units: psi * 100 Limit in PSI before warning is triggered
30	INT16	PeakPressure_ErrorLimit	PeakPressure_ErrorLimit Min: 50 Max: 1000 Default: 150 Units: psi * 100 Limit in PSI before error is triggered
32	UINT16	PeakPressure_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Maximum Pressure test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
34	UINT16	PeakPressure_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits

36	INT16	AveragePressure_WarnLimit	AveragePressure_WarnLimit Min: 50 Max: 1000 Default: 100 Units: psi * 100 Limit in PSI before warning is triggered
38	INT16	AveragePressure_ErrorLimit	AveragePressure_ErrorLimit Min: 50 Max: 1000 Default: 150 Units: psi * 100 Limit in PSI before error is triggered
40	UINT16	AveragePressure_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Average Pressure test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
42	UINT16	AveragePressure_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
44	INT16	CutoffPressure_WarnLimit	CutoffPressure_WarnLimit Min: 50 Max: 1000 Default: 100 Units: psi * 100 Limit in PSI before warning is triggered
46	INT16	CutoffPressure_ErrorLimit	CutoffPressure_ErrorLimit Min: 50 Max: 1000 Default: 150 Units: psi * 100 Limit in PSI before error is triggered
48	UINT16	CutoffPressure_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Cutoff Pressure test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.

50	UINT16	CutoffPressure_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
52	UINT16	AverageRate_WarnLimit	AverageRate_WarnLimit Min: 100 Max: 1000 Default: 250 Units: ml/s * 1000 Limit in rate before warning is tiggered
54	UINT16	AverageRate_ErrorLimit	AverageRate_ErrorLimit Min: 100 Max: 1000 Default: 500 Units: ml/s * 1000 Limit in rate before error is tiggered
56	UINT16	AverageRate_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Average Motor Rate test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
58	UINT16	AverageRate_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
60	UINT16	TotalVolume_WarnLimit	TotalVolume_WarnLimit Min: 50 Max: 5000 Default: 250 Units: ml * 1000 Upper volume limit before warning is triggered
62	UINT16	TotalVolume_ErrorLimit	TotalVolume_ErrorLimit Min: 50 Max: 5000 Default: 500 Units: ml * 1000 Upper volume limit before error is triggered

64	UINT16	TotalVolume_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Total Motor Volume test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
66	UINT16	TotalVolume_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
68	INT16	AverageCurrent_WarnLimit	AverageCurrent_WarnLimit Min: 50 Max: 1000 Default: 100 Units: mA Limit in PSI before warning is triggered
70	INT16	AverageCurrent_ErrorLimit	AverageCurrent_ErrorLimit Min: 50 Max: 1000 Default: 150 Units: mA Limit in PSI before error is triggered
72	UINT16	AverageCurrent_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Average Pressure test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
74	UINT16	AverageCurrent_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
76	INT16	PredictedCurrent_WarnLimit	PredictedCurrent_WarnLimit Min: 50 Max: 1000 Default: 100 Units: mA Limit in PSI before warning is triggered

78	INT16	PredictedCurrent_ErrorLimit	PredictedCurrent_ErrorLimit Min: 50 Max: 1000 Default: 150 Units: mA Limit in PSI before error is triggered
80	UINT16	PredictedCurrent_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Average Pressure test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
82	UINT16	PredictedCurrent_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
84	UINT16	AirDetect_WarnLimit	AirDetect_WarnLimit Min: 100 Max: 2000 Default: 250 Units: ml * 1000 Limit in air volume before warning is triggered
86	UINT16	AirDetect_ErrorLimit	AirDetect_ErrorLimit Min: 100 Max: 2000 Default: 500 Units: ml * 1000 Limit in air volume before error is triggered
88	UINT16	AirDetect_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Air Detect Volume test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
90	UINT16	AirDetect_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits

92	INT16	DisTemp_WarnLimit	DisTemp_WarnLimit Min: 2000 Max: 9000 Default: 6000 Units: C * 100 Temperature limit before warning is triggered
94	INT16	DisTemp_ErrorLimit	DisTemp_ErrorLimit Min: 2000 Max: 9000 Default: 6000 Units: C * 100 Temperature limit before error is triggered
96	UINT16	DisTemp_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Air Detect Volume test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
98	UINT16	DisTemp_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
100	INT16	AverageDeltaPressure_WarnLimit	AverageDeltaPressure_WarnLimit Min: 50 Max: 1000 Default: 100 Units: psi * 100 Limit in PSI before warning is triggered
102	INT16	AverageDeltaPressure_ErrorLimit	AverageDeltaPressure_ErrorLimit Min: 50 Max: 1000 Default: 150 Units: psi * 100 Limit in PSI before error is triggered
104	UINT16	AverageDeltaPressure_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Average Pressure test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.

106	UINT16	AverageDeltaPressure_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
108	INT16	MaxFiltrationPrs_WarnLimit	MaxFiltrationPrs_WarnLimit Min: 100 Max: 3000 Default: 1500 Units: psi * 100 Limit in PSI before warning is triggered
110	INT16	MaxFiltrationPrs_ErrorLimit	MaxFiltrationPrs_ErrorLimit Min: 100 Max: 3000 Default: 1800 Units: psi * 100 Limit in PSI before error is triggered
112	UINT16	MaxFiltrationPrs_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Filtration - Maximum Upstream Pressure test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
114	UINT16	MaxFiltrationPrs_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
116	UINT16	FiltrationRate_WarnLimit	FiltrationRate_WarnLimit Min: 100 Max: 1000 Default: 300 Units: ml/s * 1000 Limit in rate before warning is triggered
118	UINT16	FiltrationRate_ErrorLimit	FiltrationRate_ErrorLimit Min: 100 Max: 1000 Default: 500 Units: ml/s * 1000 Limit in rate before error is triggered

120	UINT16	FiltrationRate_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Fill - Average Motor Rate test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
122	UINT16	FiltrationRate_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
124	UINT16	FiltrationTime_WarnLimit	FiltrationTime_WarnLimit Min: 20 Max: 300 Default: 50 Units: s * 100 Limit in time before warning is tiggered
126	UINT16	FiltrationTime_ErrorLimit	FiltrationTime_ErrorLimit Min: 20 Max: 300 Default: 60 Units: s * 100 Limit in time before error is tiggered
128	UINT16	FiltrationTime_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Fill - Time test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
130	UINT16	FiltrationTime_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
132	INT16	MinFillPrs_WarnLimit	MinFillPrs_WarnLimit Min: -1400 Max: 100 Default: "-900" Units: psi * 100 Limit in PSI before warning is triggered

134	INT16	MinFillPrs_ErrorLimit	MinFillPrs_ErrorLimit Min: -1400 Max: 100 Default: "-1000" Units: psi * 100 Limit in PSI before error is triggered
136	UINT16	MinFillPrs_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Fill - Minimum Fill Pressure test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
138	UINT16	MinFillPrs_TestType	test type greater then or less then Min: 1 Max: 1 Default: 1 Test if the value is LESS than the alarm limits
140	UINT16	AverageFillRate_WarnLimit	AverageFillRate_WarnLimit Min: 100 Max: 1500 Default: 250 Units: ml/s * 1000 Limit in rate before warning is triggered
142	UINT16	AverageFillRate_ErrorLimit	AverageFillRate_ErrorLimit Min: 100 Max: 1500 Default: 500 Units: ml/s * 1000 Limit in rate before error is triggered
144	UINT16	AverageFillRate_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Fill - Average Motor Rate test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
146	UINT16	AverageFillRate_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits

148	UINT16	FillTime_WarnLimit	FillTime_WarnLimit Min: 20 Max: 2000 Default: 50 Units: s * 100 Limit in time before warning is triggered
150	UINT16	FillTime_ErrorLimit	FillTime_ErrorLimit Min: 20 Max: 2000 Default: 75 Units: s * 100 Limit in time before error is triggered
152	UINT16	FillTime_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Fill - Time test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
154	UINT16	FillTime_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
156	UINT16	FillMotorMisMatch_WarnLimit	FillMotorMisMatch_WarnLimit Min: 50 Max: 5000 Default: 100 Units: ml * 1000 Upper volume limit before warning is triggered
158	UINT16	FillMotorMisMatch_ErrorLimit	FillMotorMisMatch_ErrorLimit Min: 50 Max: 5000 Default: 250 Units: ml * 1000 Upper volume limit before error is triggered
160	UINT16	FillMotorMisMatch_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Total Motor Volume test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.

162	UINT16	FillMotorMisMatch_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
164	UINT16	CycleTime_WarnLimit	CycleTime_WarnLimit Min: 20 Max: 2000 Default: 50 Units: s * 100 Limit in time before warning is tiggered
166	UINT16	CycleTime_ErrorLimit	CycleTime_ErrorLimit Min: 20 Max: 2000 Default: 75 Units: s * 100 Limit in time before error is tiggered
168	UINT16	CycleTime_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Fill - Time test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
170	UINT16	CycleTime_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
172	INT16	MinVac_WarnLimit	MinVac_WarnLimit Min: -1500 Max: -600 Default: "-1000" Units: psi * 100 Limit in PSI before warning is triggered for low utility vacuum
174	INT16	MinVac_ErrorLimit	MinVac_ErrorLimit Min: -1500 Max: -600 Default: "-800" Units: psi * 100 Limit in PSI before error is triggered for low utility vacuum

176	UINT16	MinVac_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Cycle - Minimum utility vacuum test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
178	UINT16	MinVac_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is LESS than the alarm limits
180	INT16	MinN2_WarnLimit	MinN2_WarnLimit Min: 3000 Max: 8000 Default: 3800 Units: psi * 100 Limit in PSI before warning is triggered for low utility N2
182	INT16	MinN2_ErrorLimit	MinN2_ErrorLimit Min: 3000 Max: 8000 Default: 3800 Units: psi * 100 Limit in PSI before error is triggered for low utility N2
184	UINT16	MinN2_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Cycle - Minimum utility N2 test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
186	UINT16	MinN2_TestType	test type greater then or less then Min: 1 Max: 1 Default: 1 Test if the value is LESS than the alarm limits
188	INT16	ReadyCorrection_WarnLimit	ReadyCorrection_WarnLimit Min: 100 Max: 200 Default: 200 Units: ml * 1000 Upper volume limit before warning is triggered

190 INT16	ReadyCorrection_ErrorLimit	ReadyCorrection_ErrorLimit Min: 100 Max: 200 Default: 200 Units: ml * 1000 Upper volume limit before error is triggered
192 UINT16	ReadyCorrection_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Ready - Ready Correction Volume test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
194 UINT16	ReadyCorrection_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
196 UINT16	FlowVolume_WarnLimit	FlowVolume_WarnLimit Min: 50 Max: 5000 Default: 250 Units: ml * 1000 Upper flow volume limit before warning is triggered
198 UINT16	FlowVolume_ErrorLimit	FlowVolume_ErrorLimit Min: 50 Max: 5000 Default: 500 Units: ml * 1000 Upper flow volume limit before error is triggered
200 UINT16	FlowVolume_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Total Flow Volume test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
202 UINT16	FlowVolume_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits

204	UINT16	VentTime_WarnLimit	VentTime_WarnLimit Min: 10 Max: 300 Default: 20 Units: s * 100 Limit in time before warning is triggered
206	UINT16	VentTime_ErrorLimit	VentTime_ErrorLimit Min: 10 Max: 300 Default: 40 Units: s * 100 Limit in time before error is triggered
208	UINT16	VentTime_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Vent - Time test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
210	UINT16	VentTime_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
212	UINT16	reservedTest4_WarnLimit	reservedTest4_WarnLimit Default: 0
214	UINT16	reservedTest4_ErrorLimit	reservedTest4_ErrorLimit Default: 0
216	UINT16	reservedTest4_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3
218	UINT16	reservedTest4_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2
220	UINT16	PeakPressureTime_WarnLimit	PeakPressureTime_WarnLimit Default: 40 Units: ms Limit in mSeconds before warning is triggered
222	UINT16	PeakPressureTime_ErrorLimit	PeakPressureTime_ErrorLimit Default: 60 Units: ms Limit in mSeconds before error is triggered

224	UINT16	PeakPressureTime_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Peak pressure test - Time test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
226	UINT16	PeakPressureTime_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2 Test if the value is GREATER than the alarm limits
228	UINT16	ConfirmationMiniStyle_WarnLimit	ConfirmationMiniStyle_WarnLimit Min: 10 Max: 99 Default: 85 Units: % Limit in percent before warning is triggered
230	UINT16	ConfirmationMiniStyle_ErrorLimit	ConfirmationMiniStyle_ErrorLimit Min: 10 Max: 99 Default: 80 Units: % Limit in percent before error is triggered
232	UINT16	ConfirmationMiniStyle_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3 Disable alarms or select between warning/error for the Dispense - Pressure Profile Compare (IGMINI style) test. The hide selection will disable the alarm and hide the test from this page. Click show all to show the test again to enable.
234	UINT16	ConfirmationMiniStyle_TestType	test type greater then or less then Min: 1 Max: 1 Default: 1 Test if the value is LESS than the alarm limits
236	UINT16	reservedTest7_WarnLimit	reservedTest7_WarnLimit Default: 0
238	UINT16	reservedTest7_ErrorLimit	reservedTest7_ErrorLimit Default: 0
240	UINT16	reservedTest7_Enable	Disable, Error, Warn, Both Max: 3 Default: 0 Values: Disabled=0, Warning=1, Error=2, Hidden=3

242	UINT16	reservedTest7_TestType	test type greater then or less then Min: 2 Max: 2 Default: 2
244	ULONG	Time	Time Written Default: 0
248	ULONG	Cycle	Cycle when written Default: 0

DC_STATUS: DC Status Size: 124 bytes

Dispense confirmation results for the most recent or reference cycles.

Command Code: 158 DC_STATUS	
Send Data	None
Receive Data	124 bytes data as defined below
Access Level	ENGINEER

DC Status Structure		Size: 124 bytes
Type	Field	Description
0 UINT16	RecipeNum	Recipe number Default: 0 This selects the DC limits for the recipe.
2 UINT16	reserved0	reserved0 Default: 0
4 ULONG	Cycle	Cycle Number Default: 0
8 ULONG	ReferenceCycle	Reference Cycle Number Default: 0
12 INT16	ReadyPressure_VUT	Value under test Default: 0 Units: psi * 100 Displays the absolute difference between the last dispense cycle and the reference cycle
14 UINT16	ReadyPressure_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
16 INT16	ReadyCorrection_VUT	Value under test Default: 0 Units: ml * 1000 Displays the absolute value from the last dispense cycle directly
18 UINT16	ReadyCorrection_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
20 UINT16	Confirmation_VUT	Value under test Default: 0 Units: % Displays the value from the last dispense cycle directly
22 UINT16	Confirmation_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3

24	UINT16	FlowConfirmation_VUT	Value under test Default: 0 Units: % Displays the value from the last dispense cycle directly
26	UINT16	FlowConfirmation_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
28	INT16	PeakPressure_VUT	Value under test Default: 0 Units: psi * 100 Displays the absolute difference between the last dispense cycle and the reference cycle
30	UINT16	PeakPressure_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
32	INT16	AveragePressure_VUT	Value under test Default: 0 Units: psi * 100 Displays the absolute difference between the last dispense cycle and the reference cycle
34	UINT16	AveragePressure_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
36	INT16	CutoffPressure_VUT	Value under test Default: 0 Units: psi * 100 Displays the absolute difference between the last dispense cycle and the reference cycle
38	UINT16	CutoffPressure_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
40	UINT16	AverageRate_VUT	Value under test Default: 0 Units: ml/s * 1000 Displays the absolute difference between the last dispense cycle and the reference cycle
42	UINT16	AverageRate_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
44	UINT16	TotalVolume_VUT	Value under test Default: 0 Units: ml * 1000 Displays the absolute difference between the last dispense cycle and the reference cycle
46	UINT16	TotalVolume_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3

48	INT16	AverageCurrent_VUT	Value under test Default: 0 Units: mA Displays the absolute difference between the last dispense cycle and the reference cycle
50	UINT16	AverageCurrent_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
52	INT16	PredictedCurrent_VUT	Value under test Default: 0 Units: mA Displays the absolute difference between the last dispense cycle and the reference cycle
54	UINT16	PredictedCurrent_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
56	UINT16	AirDetect_VUT	Value under test Default: 0 Units: ml * 1000 Displays the value from the last dispense cycle directly
58	UINT16	AirDetect_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
60	INT16	DisTemp_VUT	Value under test Default: 0 Units: C * 100 Displays the value from the last dispense cycle directly
62	UINT16	DisTemp_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
64	INT16	AverageDeltaPressure_VUT	Value under test Default: 0 Units: psi * 100 Displays the absolute difference between the last dispense cycle and the reference cycle
66	UINT16	AverageDeltaPressure_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
68	INT16	MaxFiltrationPrs_VUT	Value under test Default: 0 Units: psi * 100 Displays the value from the last dispense cycle directly
70	UINT16	MaxFiltrationPrs_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3

72	UINT16	FiltrationRate_VUT	Value under test Default: 0 Units: ml/s * 1000 Displays the absolute difference between the last dispense cycle and the reference cycle
74	UINT16	FiltrationRate_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
76	UINT16	FiltrationTime_VUT	Value under test Default: 0 Units: sec * 100 Displays the absolute difference between the last dispense cycle and the reference cycle
78	UINT16	FiltrationTime_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
80	INT16	MinFillPrs_VUT	Value under test Default: 0 Units: psi * 100 Displays the value from the last dispense cycle directly
82	UINT16	MinFillPrs_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
84	UINT16	AverageFillRate_VUT	Value under test Default: 0 Units: ml/s * 1000 Displays the absolute difference between the last dispense cycle and the reference cycle
86	UINT16	AverageFillRate_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
88	UINT16	FillTime_VUT	Value under test Default: 0 Units: s * 100 Displays the absolute difference between the last dispense cycle and the reference cycle
90	UINT16	FillTime_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
92	INT16	FillMotorMisMatch_VUT	Value under test Default: 0 Units: ml * 1000 Displays the value from the last dispense cycle directly
94	UINT16	FillMotorMisMatch_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3

96	UINT16	CycleTime_VUT	Value under test Default: 0 Units: s * 100 Displays the absolute difference between the last dispense cycle and the reference cycle
98	UINT16	CycleTime_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
100	INT16	MinVac_VUT	Value under test Default: 0 Units: psi * 100 Displays the value from the last dispense cycle directly
102	UINT16	MinVac_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
104	INT16	MinN2_VUT	Value under test Default: 0 Units: psi * 100 Displays the value from the last dispense cycle directly
106	UINT16	MinN2_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
108	UINT16	FlowVolume_VUT	Value under test Default: 0 Units: ml * 1000 Displays the absolute difference between the last dispense cycle and the reference cycle
110	UINT16	FlowVolume_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
112	UINT16	VentTime_VUT	Value under test Default: 0 Units: s * 100 Displays the absolute difference between the last dispense cycle and the reference cycle
114	UINT16	VentTime_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3
116	UINT16	PeakPressureTime_VUT	Value under test Default: 0 Units: ms Displays the absolute difference between the last dispense cycle and the reference cycle
118	UINT16	PeakPressureTime_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3

120	UINT16	ConfirmationMiniStyle_VUT	Value under test Default: 0 Units: % Displays the value from the last dispense cycle directly
122	UINT16	ConfirmationMiniStyle_Status	Status of the test: warning, error, or none. Default: 0 Values: None=0, Warning=2, Error=3

DISPENSE_INFO: Dispense Information Size: 164 bytes

Dispense data for saved reference dispenses (by recipe number) or the most recent dispense (send 0 for recipe).

Command Code: 155 DISPENSE_INFO	
Send Data	2 bytes for Recipe Number
Receive Data	164 bytes data as defined below
Access Level	ENGINEER

Dispense Information Structure		Size: 164 bytes
Type	Field	Description
0 UINT16	RecipeNum	Recipe number Max: 12 Default: 1 This selects the DC limits for the recipe.
2 UINT16	reserved0	reserved0 Default: 0
4 ULONG	Cycle	Cycle Number Default: 0
8 ULONG	ReferenceCycle	Reference Cycle Number Default: 0
12 ULONG	Time	Time of dispense (trigger) Default: 0
16 STRING	JobInfo[24]	Job Information Default: 0 Job Identification information for this dispense
40 UINT16	SlotID	Slot ID Default: 0 This was the slot ID of this dispense
42 UINT16	Reserved	Reserved Default: 0
44 INT16	ReadyPressure	Ready - Ready Pressure Default: 0 Units: psi * 100 Displays the average ready pressure right before last dispense cycle.
46 INT16	ReferenceReadyPressure	Reference Ready - Ready Pressure Default: 0 Units: psi * 100 Displays the average ready pressure right before reference dispense cycle.

48	INT16	ReadyCorrection	Ready - Ready Correction Volume Default: 0 Units: ml * 1000 Dispense the amount of volume that was used to maintain the ready pressure. This is the value of correction at the start of the last cycle.
50	INT16	ReferenceReadyCorrection	Reference Ready - Ready Correction Volume Default: 0 Units: ml * 1000 Dispense the amount of volume that was used to maintain the ready pressure. This is the value of correction at the start of the reference cycle.
52	UINT16	Confirmation	Dispense - Pressure Profile Compare Default: 0 Units: % Displays the percent of fit between the pressure profiles. Compares the last dispense cycle to the reference dispense cycle of that recipe.
54	UINT16	ReferenceConfirmation	Reference Dispense - Pressure Profile Compare Default: 0 Units: % Displays the percent of fit between the pressure profiles. By default this will always be 100%
56	UINT16	FlowConfirmation	Dispense - Flow Profile Compare Default: 0 Units: % Displays the percent of fit between the flow profiles. Compares the last dispense cycle to the reference dispense cycle of that recipe.
58	UINT16	ReferenceFlowConfirmation	Reference Dispense - Flow Profile Compare Default: 0 Units: % Displays the percent of fit between the flow profiles. By default this will always be 100%
60	INT16	PeakPressure	Dispense - Maximum Pressure Default: 0 Units: psi * 100 Displays the maximum dispense pressure of the last dispense cycle. The highest pressure value recorded during any of the dispense segments of that cycle.
62	INT16	ReferencePeakPressure	Reference Dispense - Maximum Pressure Default: 0 Units: psi * 100 Displays the maximum dispense pressure of the reference dispense cycle. The highest pressure value recorded during any of the dispense segments of that cycle.

64	INT16	AveragePressure	Dispense - Average Pressure Default: 0 Units: psi * 100 Displays the average dispense pressure of the last dispense cycle. The average pressure value calculated during all of the dispense segments of that cycle.
66	INT16	ReferenceAveragePressure	Reference Dispense - Average Pressure Default: 0 Units: psi * 100 Displays the average dispense pressure of the reference dispense cycle. The average pressure value calculated during all of the dispense segments of that cycle.
68	INT16	CutoffPressure	Dispense - Cutoff Pressure Default: 0 Units: psi * 100 Displays the cutoff pressure of the last dispense cycle. The cutoff pressure value is calculated using the average pressure value taken after the outlet valve closes, post dispense segment.
70	INT16	ReferenceCutoffPressure	Reference Dispense - Cutoff Pressure Default: 0 Units: psi * 100 Displays the cutoff pressure of the reference dispense cycle. The cutoff pressure value is calculated using the average pressure value taken after the outlet valve closes, post dispense segment.
72	UINT16	AverageRate	Dispense - Average Motor Rate Default: 0 Units: ml/s * 1000 Displays the average dispense rate of the last dispense cycle. The average rate value calculated during all of the dispense segments of that cycle.
74	UINT16	ReferenceAverageRate	Reference Dispense - Average Motor Rate Default: 0 Units: ml/s * 1000 Displays the average dispense rate of the reference dispense cycle. The average rate value calculated during all of the dispense segments of that cycle.
76	UINT16	TotalVolume	Dispense - Total Motor Volume Default: 0 Units: ml * 1000 Displays the total volume of the last dispense cycle. The total volume is calculated across all of the dispense segments of that cycle.

78	UINT16	ReferenceTotalVolume	Reference Dispense - Total Motor Volume Default: 0 Units: ml * 1000 Displays the total volume of the reference dispense cycle. The total volume is calculated across all of the dispense segments of that cycle.
80	INT16	AverageCurrent	Dispense - Average Motor Current Default: 0 Units: mA Displays the average dispense motor current of the last dispense cycle. The average motor current value calculated during all of the dispense segments of that cycle.
82	INT16	ReferenceAverageCurrent	Reference Dispense - Average Motor Current Default: 0 Units: mA Displays the average dispense motor current of the reference dispense cycle. The average motor current value calculated during all of the dispense segments of that cycle.
84	INT16	PredictedCurrent	Dispense - Predicted Motor Current Default: 0 Units: mA Displays the predicted dispense motor current of the last dispense cycle. The predicted motor current value calculated during all of the dispense segments of that cycle.
86	INT16	ReferencePredictedCurrent	Reference Dispense - Predicted Motor Current Default: 0 Units: mA Displays the predicted dispense motor current of the reference dispense cycle. The predicted motor current value calculated during all of the dispense segments of that cycle.
88	UINT16	AirDetect	Dispense - Air Detect Volume Default: 0 Units: ml * 1000 Displays the calculated dispense air detection value for the last dispense cycle. The dispense air detection value is calculated during the recharge of that cycle
90	UINT16	ReferenceAirDetect	Reference Dispense - Air Detect Volume Default: 0 Units: ml * 1000 Displays the calculated dispense air detection value for the reference dispense cycle. The dispense air detection value is calculated during the recharge of that cycle
92	INT16	DisTemp	Dispense - Temperature Default: 0 Units: C * 100 Displays the value of the temperature reading of the dispense pressure sensor. The value was recorded at the end of the last dispense segment

94 INT16	ReferenceDisTemp	Reference Dispense - Temperature Default: 0 Units: C * 100 Displays the value of the temperature reading of the dispense pressure sensor. The value was recorded at the end of the reference dispense segment
96 UINT16	FlowVolume	Dispense - Total Flow Volume Default: 0 Units: ml * 1000 Displays the total flow volume of the last dispense cycle. The total flow volume is calculated using nozzle flow sensor. It is calculated across all of the dispense segments of that cycle.
98 UINT16	ReferenceFlowVolume	Reference Dispense - Total Flow Volume Default: 0 Units: ml * 1000 Displays the total flow volume of the reference dispense cycle. The total flow volume is calculated using nozzle flow sensor. It is calculated across all of the dispense segments of that cycle.
100 INT16	AverageDeltaPressure	Filtration - Average Delta Pressure Default: 0 Units: psi * 100 Displays the average filtration delta pressure of the last dispense cycle.
102 INT16	ReferenceAverageDeltaPressure	Reference Filtration - Average Delta Pressure Default: 0 Units: psi * 100 Displays the average filtration delta pressure of the reference dispense cycle.
104 INT16	MaxFiltrationPrs	Filtration - Maximum Upstream Pressure Default: 0 Units: psi * 100 Displays the maximum downstream filtration pressure of the last dispense cycle. The highest pressure value recorded during any of part of the filtration segment of that cycle.
106 INT16	ReferenceMaxFiltrationPrs	Reference Filtration - Maximum Upstream Pressure Default: 0 Units: psi * 100 Displays the maximum downstream filtration pressure of the reference dispense cycle. The highest pressure value recorded during any of part of the filtration segment of that cycle.
108 UINT16	FiltrationRate	Filtration - Average Motor Rate Default: 0 Units: ml/s * 1000 Displays the average filtration rate of the last dispense cycle.

110 UINT16	ReferenceFiltrationRate	Reference Filtration - Average Motor Rate Default: 0 Units: ml/s * 1000 Displays the average filtration rate of the reference dispense cycle.
112 UINT16	FiltrationTime	Filtration - Total Time Default: 0 Units: s * 100 Displays the filtration time of the last dispense cycle.
114 UINT16	ReferenceFiltrationTime	Reference Filtration - Total Time Default: 0 Units: s * 100 Displays the filtration time of the reference dispense cycle.
116 UINT16	VentTime	Vent - Total Time Default: 0 Units: s * 100 Displays the Vent time of the last dispense cycle.
118 UINT16	ReferenceVentTime	Reference Vent - Total Time Default: 0 Units: s * 100 Displays the Vent time of the reference dispense cycle.
120 INT16	MinFillPrs	Fill - Minimum Fill Pressure Default: 0 Units: psi * 100 Displays the minimum fill pressure of the last dispense cycle. The lowest pressure value recorded during any of part of the fill segment of that cycle.
122 INT16	ReferenceMinFillPrs	Reference Fill - Minimum Fill Pressure Default: 0 Units: psi * 100 Displays the minimum fill pressure of the reference dispense cycle. The lowest pressure value recorded during any of part of the fill segment of that cycle.
124 UINT16	AverageFillRate	Fill - Average Motor Rate Default: 0 Units: ml/s * 1000 Displays the average fill rate of the last dispense cycle. The average rate value calculated during all of the dispense segments of that cycle.
126 UINT16	ReferenceAverageFillRate	Reference Fill - Average Motor Rate Default: 0 Units: ml/s * 1000 Displays the average fill rate of the reference dispense cycle. The average rate value calculated during all of the dispense segments of that cycle.

128	UINT16	FillTime	Fill - Total Time Default: 0 Units: s * 100 Displays the fill time of the last dispense cycle.
130	UINT16	ReferenceFillTime	Reference Fill - Total Time Default: 0 Units: s * 100 Displays the fill time of the reference dispense cycle.
132	UINT16	reservedTest4	reserved future use Default: 0
134	UINT16	ReferencereservedTest4	reserved future use Default: 0
136	INT16	FillMotorMisMatch	Cycle - Fill Motor Mismatch Default: 0 Units: ml * 1000 Displays the volume of the fill motor mismatch for the last dispense cycle.
138	INT16	ReferenceFillMotorMisMatch	Cycle Dispense - Fill Motor Mismatch Default: 0 Units: ml * 1000 Displays the volume of the fill motor mismatch for the reference dispense cycle.
140	UINT16	CycleTime	Cycle - Total Time Default: 0 Units: s * 100 Displays the total cycle time of the last dispense cycle.
142	UINT16	ReferenceCycleTime	Reference Cycle - Total Time Default: 0 Units: s * 100 Displays the total cycle time of the reference dispense cycle.
144	INT16	MinVac	Cycle - Minimum Utility Vacuum Default: 0 Units: psi * 100 Displays the minimum utility vacuum pressure of the last dispense cycle. The lowest vacuum value recorded during any of part of that cycle.
146	INT16	ReferenceMinVac	Reference Cycle - Minimum Utility Vacuum Default: 0 Units: psi * 100 Displays the minimum utility vacuum pressure of the reference dispense cycle. The lowest vacuum value recorded during any of part of that cycle.

148	INT16	MinN2	Cycle - Minimum Utility N2 Default: 0 Units: psi * 100 Displays the minimum utility N2 pressure of the last dispense cycle. The lowest N2 value recorded during any of part of that cycle.
150	INT16	ReferenceMinN2	Reference Cycle - Minimum Utility N2 Default: 0 Units: psi * 100 Displays the minimum utility N2 pressure of the reference dispense cycle. The lowest N2 value recorded during any of part of that cycle.
152	UINT16	PeakPressureTime	Dispense - Max Pressure Time Default: 0 Units: ms Displays the time during the last dispense at which the maximum pressure occur.
154	UINT16	ReferencePeakPressureTime	Reference Dispense - Max Pressure Time Default: 0 Units: ms Displays the time during the reference dispense at which the maximum pressure occur.
156	UINT16	ConfirmationMiniStyle	Dispense - Pres Profile Cmp (IGMINI style) Default: 0 Units: % Displays the percent of fit between the pressure profiles (IGMINI style). Compares the last dispense cycle to the reference dispense cycle of that recipe. This version matches IGMINI start and endpoints for the comparison
158	UINT16	ReferenceConfirmationMiniStyle	Reference Dispense - Pres Profile Cmp (IGMINI style) Default: 0 Units: % Displays the percent of fit between the pressure profiles (IGMINI style). By default this will always be 100%
160	UINT16	reservedTest7	reserved future use Default: 0
162	UINT16	ReferencereservedTest7	reserved future use Default: 0

SAVE_REFERENCE: Save Reference Size: 28 bytes

Save the current cycle as the reference for the recipe.

Command Code: 156 SAVE_REFERENCE	
Send Data	2 bytes for Recipe Number
Receive Data	28 bytes data as defined below
Access Level	ENGINEER

Save Reference Structure		Size: 28 bytes
Type	Field	Description
0 UINT16	RecipeNum	Recipe to Save To Min: 1 Max: 12 Default: 0
2 UINT16	DeleteReference	(delete the reference) Default: 0
4 STRING	ReferenceTitle[24]	Description Only used for non recipe (partial emulation) Default: 0 Save the name associated with this reference

JOB_INFO: Job Information Size: 28 bytes

Used to write a string and a number to identify the NEXT dispense. This information is then copied into the DISPENSE_INFO for that dispense cycle.

Command Code: 157 WRITE JOB_INFO	
Send Data	28 bytes data as defined below
Receive Data	None
Access Level	NONE

Job Information Structure			Size: 28 bytes
Type	Field	Description	
0 STRING	JobInfo[24]	Job Information Default: 0 Job Identification information for next dispense	
24 UINT16	SlotID	Slot ID Default: 0 This is the slot ID of the next dispense	
26 UINT16	RecipeNum	Recipe Default: 0 Expected recipe number	

START_VENTONLY: Vent only start

This command will perform a vent on the system.

Command Code: 160 START_VENTONLY	
Send Data	None
Receive Data	None
Access Level	OPERATOR

PASSWORDS: Passwords Size: 16 bytes

This command is used to set-up and maintain the system passwords. See the [Appendix](#) for more detailed information about this command.

Command Code: 162 PASSWORDS	
Send Data	<i>None</i>
Receive Data	16 bytes data as defined below
Access Level	<i>NONE</i>

Command Code: 161 WRITE PASSWORDS	
Send Data	16 bytes data as defined below
Receive Data	<i>None</i>
Access Level	<i>NONE</i>

Passwords Structure			Size: 16 bytes
Type	Field	Description	
0 UINT16	Enable	Require access password Default: 0 Values: Disable=0, Enable=1	
2 UINT16	Customer	Customer behavior number Default: 0	
4 UINT16	Operator	Operator level password (key) Default: 31478	
6 UINT16	Engineer	Engineer level password (key) Default: 21944	
8 UINT16	Owner	Owner level password (key) Default: 53892	
10 UINT16	reserved0	reserved Default: 0	
12 ULONG	reserved1	reserved Default: 0	

SKIPSTEPINFO: Get SkipStep info Size: 16 bytes

Command Code: 166 SKIPSTEPINFO	
Send Data	None
Receive Data	16 bytes data as defined below
Access Level	NONE

Get SkipStep info Structure			Size: 16 bytes
Type	Field	Description	
0 ULONG	Cycle	Cycle for this information Default: 0	
4 INT16	StepMismatch_uL	StepMismatch uL Default: 0 Units: ul	
6 INT16	reserved1	reserved Default: 0	
8 ULONG	reserved2	reserved Default: 0	
12 ULONG	reserved3	reserved Default: 0	

LAST_DIS_INFO_2: Last Dispense Info 2 Size: 16 bytes

Command Code: 167 LAST_DIS_INFO_2	
Send Data	None
Receive Data	16 bytes data as defined below
Access Level	NONE

Last Dispense Info 2 Structure			Size: 16 bytes
Type	Field	Description	
0 ULONG	Cycle	Cycle for this information Default: 0	
4 ULONG	DispenseVolume	DispenseVolume Default: 0 Units: ml * 1000	
8 ULONG	reserved1	reserved Default: 0	
12 ULONG	reserved2	reserved Default: 0	

START_FLUSH_PUMP: Start Flush Pump Size: 40 bytes

This command will start the pump flush function. This is designed to run the motors while the pump is being flushed with solvent.

Command Code: 170 START_FLUSH_PUMP	
Send Data	None
Receive Data	40 bytes data as defined below
Access Level	NONE

Command Code: 171 WRITE START_FLUSH_PUMP	
Send Data	40 bytes data as defined below
Receive Data	None
Access Level	NONE

Start Flush Pump Structure			Size: 40 bytes
Type	Field	Description	
0 ULONG	FlushFilterRate	Filter Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the rate for the filtration segment of the flush pump procedure.	
4 ULONG	Reserved1	Reserved1 Default: 0	
8 ULONG	FlushPurgeRate	Purge Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the rate for the purge segment of the flush pump procedure.	
12 ULONG	FlushPurgeFreq	Purge Frequency Max: 1000 Default: 10 Units: cycles This sets the frequency of the purge segment during the flush pump procedure.	

16	ULONG	FlushVentRate	Vent Rate Min: 100 Max: 3000 Default: 500 Units: ml/s * 1000 This sets the rate for the vent segment of the flush pump procedure.
20	ULONG	FlushVentFreq	Vent Frequency Max: 1000 Default: 10 Units: cycles This sets the frequency of the vent segment during the flush pump procedure.
24	ULONG	Reserved2	Reserved2 Default: 0
28	ULONG	Reserved3	Reserved3 Default: 0
32	ULONG	Reserved4	Reserved4 Default: 0
36	ULONG	Reserved5	Reserved5 Default: 0

STOP_FLUSH_PUMP: Stop Flush Pump

This command will stop the pump flush function.

Command Code: 172 STOP_FLUSH_PUMP	
Send Data	<i>None</i>
Receive Data	<i>None</i>
Access Level	NONE

START_CYCLE_INTERNAL_VALVES: Utility: Start Cycle Internal Valves Size: 16 bytes

Start Cycle Internal Valves

Command Code: 180 START_CYCLE_INTERNAL_VALVES	
Send Data	None
Receive Data	16 bytes data as defined below
Access Level	ENGINEER

Command Code: 181 WRITE START_CYCLE_INTERNAL_VALVES	
Send Data	16 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Utility: Start Cycle Internal Valves Structure Size: 16 bytes			
Type	Field	Description	
0 ULONG	OpenWait_ms	Open duration in mSEC Min: 1 Max: 100000 Default: 1000	
4 ULONG	CloseWait_ms	Close duration in mSEC Min: 1 Max: 100000 Default: 1000	
8 ULONG	RepeatCycles	Repeat Cycles Min: 1 Max: 1000000 Default: 50	
12 ULONG	RepeatRemaining	Number of remaing until stop Default: 0	

STOP_CYCLE_INTERNAL_VALVES: Utility: Stop Cycle Internal Valves

Stop Cycle Internal Valves

Command Code: 182 STOP_CYCLE_INTERNAL_VALVES	
Send Data	None
Receive Data	None
Access Level	ENGINEER

SET_MOTORS: Utility: Set motor positions Size: 16 bytes

Set Motor Positions

Command Code: 183 SET_MOTORS	
Send Data	None
Receive Data	16 bytes data as defined below
Access Level	NONE

Command Code: 184 WRITE SET_MOTORS	
Send Data	16 bytes data as defined below
Receive Data	None
Access Level	NONE

Utility: Set motor positions Structure Size: 16 bytes			
Type	Field	Description	
0 UINT16	DispensePosition	Dispense Position Max: ST_MAX_DISPENSE_VOLUME Default: 3000 Units: mL * 1000 This sets the displacement position in ml from full volume	
2 UINT16	DispenseMotorRate	Dispense Motor Rate Max: 3000 Default: 500 Units: mL/sec * 1000 This sets the rate to move to dispense position	
4 UINT16	FillPosition	Fill Position Max: ST_MAX_DISPENSE_VOLUME Default: 3000 Units: mL * 1000 This sets the displacement position in ml from full volume	
6 UINT16	FillMotorRate	Fill Motor Rate Max: 3000 Default: 500 Units: mL/s * 1000 This sets the rate to move to fill position	
8 ULONG	Reserved1	Reserved1 Default: 0	
12 ULONG	Reserved2	Reserved2 Default: 0	

TEST_STATUS: Test Status Size: 12 bytes

Returns the status of the current or previously run diagnostic test.

Command Code: 200 TEST_STATUS	
Send Data	None
Receive Data	12 bytes data as defined below
Access Level	NONE

Test Status Structure		Size: 12 bytes
Type	Field	Description
0 UINT16	Status	Test Status Default: 0 0=Idle,1=Running
2 INT16	Complete	Percent Complete Default: 0
4 UINT16	Result	Test Result Default: 0 0=NOT RUN, 1=FAIL, 2=PASS, 3=INCOMPLETE
6 UINT16	Test	Command Code of test Default: 0
8 INT16	Cancelled	Test has been cancelled Default: 0 0=NOT CANCELLED, 1=CANCELLED
10 UINT16	reserved1	reserved Default: 0

TEST_DISBACKLASH: Self Test: Dispense Backlash Size: 12 bytes

This command will perform the Dispense Backlash self test.

Command Code: 201 TEST_DISBACKLASH	
Send Data	None
Receive Data	12 bytes data as defined below
Access Level	ENGINEER

Command Code: 202 WRITE TEST_DISBACKLASH	
Send Data	12 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Self Test: Dispense Backlash Structure			Size: 12 bytes
Type	Field	Description	
0 INT16	LowerPressureSetpoint_cPSI	Input: Lower Pressure Setpoint Min: -1200 Max: 200 Default: "-600" Units: psi * 100 Lower pressure setpoint for the beginning of the dispense motor backlash test.	
2 INT16	UpperPressureSetpoint_cPSI	Input: Upper Pressure Setpoint Min: 300 Max: 1000 Default: 600 Units: psi * 100 Upper pressure setpoint for the end of the dispense motor backlash test.	
4 INT16	BacklashUpperLimit_micrometre	Limit: Backlash Limit Default: 0 Units: um Displays the upper Pass/Fail test limit for the dispense motor backlash test.	
6 INT16	BacklashResult_micrometre	Result: Backlash Default: 0 Units: um Displays the amount of backlash detected by the dispense motor backlash test.	
8 INT16	BacklashPressureResult_cPSI	Result: Backlash Pressure Default: 0 Units: psi * 100 Displays the pressure at which the backlash was detected.	

10 UINT16	Reserved	Reserved Default: 0
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TEST_FILL_STRENGTH: Self Test: Fill Motor Strength Size: 26 bytes

This command will perform the Dry Fill Motor Strength self test.

Command Code: 203 TEST_FILL_STRENGTH	
Send Data	None
Receive Data	26 bytes data as defined below
Access Level	ENGINEER

Command Code: 204 WRITE TEST_FILL_STRENGTH	
Send Data	26 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Self Test: Fill Motor Strength Structure Size: 26 bytes			
Type	Field	Description	
0 UINT16	TestRate_uLs	Input: Test Rate Setpoint Min: 100 Max: 10000 Default: 1000 Units: ml/s * 1000 Sets the motor rate used during the test	
2 UINT16	Reserved	Reserved Default: 0	
4 LONG	PressureSetpoint_mPSI	Input: Test Pressure Setpoint Min: 15000 Max: 50000 Default: 20000 Units: psi * 1000 Sets the pressure input used for the wet motor test. Please note: This value is not required for the dry test.	
8 UINT16	CurrentSetpoint_mA	Input: Current Setpoint Max: 800 Default: 0 Units: mA Sets the motor current used during the test. If 0 is enter the system will set the current based on pressure. So the same current as normal operation.	

10	UINT16	WetOrDry	Input: Select Wet or Dry test Max: 1 Default: 0 Values: Dry Test=0, Wet Test=1 Select which the type of test Wet or Dry. A wet test requires a fully primed system, a dry test require a dry system connected to a pressure controller
12	INT16	UpperMismatchLimit_uL	Limit: Mismatch Upper Limit Default: 0 Units: ml * 1000 Displays the upper Pass/Fail test limit for the dry fill motor strength test.
14	INT16	LowerMismatchLimit_uL	Limit: Mismatch Lower Limit Default: 0 Units: ml * 1000 Displays the upper Pass/Fail test limit for the dry fill motor strength test.
16	UINT16	FullStepsPerSec	Result: Full Steps per Second Default: 0 Units: FS/s Displays the rate of the motor in Full steps per seconds. This is for reference only.
18	UINT16	TestCurrent_mA	Result: Test Current Default: 0 Units: mA Displays the current used for the strength test. Either set by the users above or set automatically by the system.
20	INT16	TestPressureResult_cPSI	Result: Test Pressure Default: 0 Units: psi * 100 Displays the pressure value for the dry fill motor strength test.
22	INT16	StartingMismatch_uL	Result: Starting Mismatch Default: 0 Units: ml * 1000 Displays the mismatch value before the test starts.
24	INT16	FinalMismatch_uL	Result: Final Mismatch Default: 0 Units: ml * 1000 Displays the mismatch value for the dry fill motor strength test.

TEST_DIS_STRENGTH: Self Test: Dry Dis Motor Strength Size: 16 bytes

This command will perform the Dry Dispense Motor Strength self test.

Command Code: 205 TEST_DIS_STRENGTH	
Send Data	None
Receive Data	16 bytes data as defined below
Access Level	ENGINEER

Command Code: 206 WRITE TEST_DIS_STRENGTH	
Send Data	16 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Self Test: Dry Dis Motor Strength Structure Size: 16 bytes			
Type	Field	Description	
0 UINT16	TestRate_uLs	Input: Test Rate Min: 100 Max: 3000 Default: 1000 Units: ml/s * 1000 Sets the motor rate used during the test	
2 INT16	UpperCurrentLimit_mA	Limit: Upper Current Limit Default: 0 Units: mA Displays the upper Pass/Fail test limit for the current of dry dispense motor strength test.	
4 INT16	PosErrorLimit_uL	Limit: Max Position Error Default: 0 Units: ml * 1000 Displays the upper Pass/Fail test limit for the position error of dispense motor strength test.	
6 INT16	PredictedTestCurrent_mA	Result: Predicted Current Default: 0 Units: mA Displays the predicted current value for the dry dispense motor strength test.	
8 INT16	TestCurrent_mA	Result: Test Current Default: 0 Units: mA Displays the current value for the dry dispense motor strength test.	

10	INT16	TestPosError_uL	Result: Test Position Error Default: 0 Units: ml * 1000 Displays the position error value for the dry dispense motor strength test.
12	INT16	TestPressure_cPSI	Result: Test Pressure Default: 0 Units: psi * 100 Displays the pressure value for the dry dispense motor strength test.
14	UINT16	AvgRate_uLs	Result: Test Rate Default: 0 Units: ml/s * 1000 Displays the average motor rate for the dry dispense motor strength test.

TEST_PRESSURE_SENSOR: Self Test: Pressure Sensor Accuracy Size: 32 bytes

This command will perform the Dry Pressure Sensor self test.

Command Code: 207 TEST_PRESSURE_SENSOR	
Send Data	None
Receive Data	32 bytes data as defined below
Access Level	ENGINEER

Command Code: 208 WRITE TEST_PRESSURE_SENSOR	
Send Data	32 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Self Test: Pressure Sensor Accuracy Structure Size: 32 bytes			
Type	Field	Description	
0 LONG	PressureSetpoint_mPSI	Input: Pressure Setpoint Min: -12000 Max: 300000 Default: 10000 Units: psi * 1000 Enter the setpoint value of the pressure controller used to run the test	
4 LONG	TestType	Input: Test Type Min: 0 Max: 2 Default: 1 Values: Dis & Fill=0, Dis Only=1, Fill Only=2 Select which pressure sensors to execute during testing	
8 FLOAT	UpperErrorLimit	Limit: Upper Error Limit Default: 0 Units: % Displays the upper Pass/Fail test limit for the full scale error of dry pressure sensor test.	
12 FLOAT	LowerErrorLimit	Limit: Upper Lower Limit Default: 0 Units: % Displays the lower Pass/Fail test limit for the full scale error of dry pressure sensor test.	
16 LONG	DisPressureResult_mPSI	Result: Dispense Pressure Default: 0 Units: psi * 1000 Displays the actual pressure reading for the dispense side pressure sensor.	

20	FLOAT	DisErrorResult_per	Result: Dispense Full Scale Err Default: 0 Units: % Displays the calculated full scale error for the dispense side pressure sensor.
24	LONG	FillPressureResult_mPSI	Result: Fill Pressure Default: 0 Units: psi * 1000 Displays the actual pressure reading for the fill side pressure sensor.
28	FLOAT	FillErrorResult_per	Result: Fill Full Scale Error Default: 0 Units: % Displays the calculated full scale error for the fill side pressure sensor.

TEST_PRESSURE_REZERO: Self Test: Pressure Rezero Size: 40 bytes

This command will perform the Dispense Pressure Sensor Rezero test.

Command Code: 216 TEST_PRESSURE_REZERO	
Send Data	None
Receive Data	40 bytes data as defined below
Access Level	ENGINEER

Command Code: 217 WRITE TEST_PRESSURE_REZERO	
Send Data	40 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Self Test: Pressure Rezero Structure		Size: 40 bytes
Type	Field	Description
0 FLOAT	UpperOffsetLimit	Limit: Upper Offset Limit Default: 0 Units: psi * 1000 Displays the upper Pass/Fail test limit for the pressure offset.
4 FLOAT	LowerOffsetLimit	Limit: Upper Offset Limit Default: 0 Units: psi * 1000 Displays the lower Pass/Fail test limit for the pressure offset.
8 LONG	PreDisPressureResult_mPSI	Result: Pre-Dispense Pressure Default: 0 Units: psi * 1000 Displays the actual pre pressure reading for the dispense stage pressure sensor.
12 FLOAT	PreDisErrorResult_per	Result: Pre-Dispense Error Default: 0 Units: % Displays the calculated pre full scale error for the dispense stage pressure sensor.
16 LONG	PreFillPressureResult_mPSI	Result: Pre-Fill Pressure Default: 0 Units: psi * 1000 Displays the actual pre pressure reading for the fill stage pressure sensor.
20 FLOAT	PreFillErrorResult_per	Result: Pre-Fill Error Default: 0 Units: % Displays the calculated pre full scale error for the fill stage pressure sensor.

24 LONG	PostDisPressureResult_mPSI	Result: Post-Dispense Pressure Default: 0 Units: psi * 1000 Displays the actual post pressure reading for the dispense stage pressure sensor.
28 FLOAT	PostDisErrorResult_per	Result: Post-Dispense Error Default: 0 Units: % Displays the calculated post full scale error for the dispense stage pressure sensor.
32 LONG	PostFillPressureResult_mPSI	Result: Post-Fill Pressure Default: 0 Units: psi * 1000 Displays the actual post pressure reading for the fill stage pressure sensor.
36 FLOAT	PostFillErrorResult_per	Result: Post-Fill Error Default: 0 Units: % Displays the calculated post full scale error for the fill stage pressure sensor.

TEST_PUMP_LEAK: Self Test: Pump Leak Test Size: 36 bytes

This command will perform the System Leak Test self test.

Command Code: 211 TEST_PUMP_LEAK	
Send Data	None
Receive Data	36 bytes data as defined below
Access Level	ENGINEER

Command Code: 212 WRITE TEST_PUMP_LEAK	
Send Data	36 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Self Test: Pump Leak Test Structure		Size: 36 bytes
Type	Field	Description
0 LONG	PressureSetpoint_mPSI	Input: Pressure Setpoint Min: -10000 Max: 300000 Default: 20000 Units: psi * 1000 Enter the value of the pressure used for the test
4 ULONG	DwellTime_Sec	Input: Dwell Time Min: 1 Max: 7500 Default: 60 Units: s Enter the value of the dwell time used to set-up the leak test
8 ULONG	TestTime_Sec	Input: Test Time Min: 1 Max: 1200 Default: 30 Units: s Enter the amount of time for the leak test
12 LONG	TestType	Input: Valve Configuration Min: 1 Max: 6 Default: 1 Values: Option 1=1, Option 2=2, Option 3=3, Option 4=4, Option 5=5, Option 6=6 Select which valve configuration you would like to test

16	LONG	LeakRateUpperLimit_mPSImin	Limit: Leak Rate Upper Limit Default: 0 Units: psi/min * 1000 Displays the upper Pass/Fail test limit for the leak rate.
20	LONG	StartingPressureResult_mPSI	Result: Starting Pressure Default: 0 Units: psi * 1000 Displays the starting pressure of the pump leak test.
24	LONG	EndingPressureResult_mPSI	Result: Ending Pressure Default: 0 Units: psi * 1000 Displays the ending pressure of the pump leak test.
28	LONG	DeltaPressureResult_mPSI	Result: Delta Pressure Default: 0 Units: psi * 1000 Displays the delta pressure of the pump leak test.
32	LONG	LeakRateResult_mPSImin	Result: Leak Rate Default: 0 Units: psi/min * 1000 Displays the leak rate result of the pump leak test.

TEST_PUMP_VALVE_FUNCTION: Self Test: Internal Valve Functional Test Size: 20 bytes

This command will perform the System Valve Function Test self test.

Command Code: 220 TEST_PUMP_VALVE_FUNCTION	
Send Data	None
Receive Data	20 bytes data as defined below
Access Level	ENGINEER

Command Code: 221 WRITE TEST_PUMP_VALVE_FUNCTION	
Send Data	20 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Self Test: Internal Valve Functional Test Size: 20 bytes			
Type	Field	Description	
0 LONG	ValveUnderTest	Input: Valve Under Test Min: 1 Max: 5 Default: 1 Values: Inlet=1, Isolate=2, Vent=3, Barrier=4, Purge=5 Select which internal valve to test	
4 LONG	DeltaPressureLimit_mPSI	Limit: Delta Pressure Min Default: 0 Units: psi/min * 1000 Displays the upper Pass/Fail test limit for the internal valve test	
8 LONG	ClosePressure_mPSI	Result: Close Pressure Default: 0 Units: psi * 1000 Displays the close pressure of the internal valve test	
12 LONG	OpenPressure_mPSI	Result: Open Pressure Default: 0 Units: psi * 1000 Displays the open pressure of the internal valve test	
16 LONG	DeltaPressureResult_mPSI	Result: Delta Pressure Default: 0 Units: psi * 1000 Displays the delta pressure of the internal valve test	

TEST_HV_VALVE_TIMING: Self Test: High Viscosity Internal Valve Timing Test Size: 28 bytes

This command will perform the High Viscosity Internal Valve Timing self test.

Command Code: 40 TEST_HV_VALVE_TIMING	
Send Data	None
Receive Data	28 bytes data as defined below
Access Level	ENGINEER

Command Code: 41 WRITE TEST_HV_VALVE_TIMING	
Send Data	28 bytes data as defined below
Receive Data	None
Access Level	ENGINEER

Self Test: High Viscosity Internal Valve Timing 28 bytes structure			
Type	Field	Description	
0 ULONG	ValveCondition	Input: VUT / Source Valves Test 1= VUT:PURGE / Source Valve(s): INLET Test 2= VUT:PURGE / Source Valve(s): VENT+ISOLATE Test 3= VUT:BARRIER / Source Valve(s): INLET+ISOLATE Test 4= VUT:BARRIER / Source Valve(s): VENT Test 5= VUT:ISOLATE / Source Valve(s): INLET+BARRIER Test 6= VUT:ISOLATE / Source Valve(s): PURGE+VENT Test 7= VUT:INLET / Source Valve(s): PURGE Test 8= VUT:INLET / Source Valve(s): BARRIER+ISOLATE Test 9= VUT:VENT / Source Valve(s): BARRIER Test 10= VUT:VENT / Source Valve(s): PURGE+ISOLATE Min: 1 Max: 10 Default: 1 Values: Test 1=1, Test 2=2, Test 3=3, Test 4=4, Test 5=5, Test 6=6, Test 7=7, Test 8=8, Test 9=9, Test 10=10 Select which Valve Under Test is used, along with which sources valves are required for this test condition	
4 ULONG	ValveTestDirection	Input: Valve Testing Direction Min: 1 Max: 2 Default: 1 Values: Open=1, Close=2 Select which direction the VUT is to be tested	
8 LONG	TargetStartPressure_mPSI	Input: Start Pressure Default: 0 Units: psi * 1000 Enter the starting pressure target of the test	

12 LONG	TargetEndPressure_mPSI	Input: End Pressure Default: 0 Units: psi * 1000 Enter the ending pressure target of the test
16 LONG	StartPressure_mPSI	Result: Start Pressure Default: 0 Units: psi * 1000 Displays the starting pressure of the test
20 LONG	EndPressure_mPSI	Result: End Pressure Default: 0 Units: psi * 1000 Displays the ending pressure target of the test
24 LONG	TestTime_ms	Result: Time from Start to End Default: 0 Units: ms Displays the time taken for pressure to decay from starting pressure to end pressure

TEST_AIR_DETECT: Self Test: Air Detect Size: 8 bytes

This command will perform the Air Detection self test.

Command Code: 214 TEST_AIR_DETECT	
Send Data	<i>None</i>
Receive Data	8 bytes data as defined below
Access Level	ENGINEER

Command Code: 215 WRITE TEST_AIR_DETECT	
Send Data	8 bytes data as defined below
Receive Data	<i>None</i>
Access Level	ENGINEER

Self Test: Air Detect Structure			Size: 8 bytes
Type	Field	Description	
0 LONG	DispenseDeltaP_mPSI	Result: Dispense Delta P Default: 0 Units: psi * 1000 Displays the pressure rise in the Dispense chamber during the test	
4 UINT16	DispenseSlope	Result: Dispense Stage Delta P Default: 0 Units: * 10 Displays the slope of the pressure rise in the dispense chamber during the test	
6 UINT16	Samples	Result: Fill Stage Air Default: 0 Displays the amount of dispense chamber pressure samples used to calculate slope	

TEST_CANCEL: Cancel Self Test

This command will cancel a self test that is running.

Command Code: 213 TEST_CANCEL	
Send Data	<i>None</i>
Receive Data	<i>None</i>
Access Level	OPERATOR

MINISTATUS: Mini Status Size: 40 bytes

Command Code: 168 MINISTATUS	
Send Data	None
Receive Data	40 bytes data as defined below
Access Level	NONE

Mini Status Structure			Size: 40 bytes
Type	Field	Description	
0 ULONG	Cycle	Cycle Default: 0	
4 INT16	Status	System Status Default: 0	
6 INT16	SystemValves	System Valves Default: 0	
8 UINT16	CriticalErrors	Counts Critical System Errors after they are logged Default: 0	
10 UINT16	Errors	Counts System Errors after they are logged Default: 0	
12 UINT16	Warnings	Counts System Warnings after they are logged Default: 0	
14 UINT16	Infos	Counts System infos after they are logged Default: 0	
16 LONG	Dis_pressure_mPSI	Dispense calibrated mPSI Default: 0 Units: psi * 1000	
20 LONG	Fill_pressure_mPSI	Fill calibrated PSI Default: 0 Units: psi * 1000	
24 UINT16	DispensePosition	Dispense Position Default: 0 Units: ml * 1000	
26 INT16	FillPosition	Fill Position Default: 0 Units: ml * 1000	
28 INT16	MotorCurrent	Motor Current Default: 0 Units: mA	
30 INT16	N2Sensor_cPSI	N2 Sensor Default: 0 Units: psi * 100	

32	INT16	VacSensor_cPSI	Vacuum Sensor Default: 0 Units: psi * 100
34	UINT16	Reserved	Reserved Default: 0
36	ULONG	NewestAlarmNum	Number of the newest alarm Default: 0

MINISTATUS2: Mini Status Small Size: 20 bytes

Command Code: 169 MINISTATUS2	
Send Data	<i>None</i>
Receive Data	20 bytes data as defined below
Access Level	NONE

Mini Status Small Structure		Size: 20 bytes
Type	Field	Description
0 INT16	Status	System Status Default: 0
2 INT16	OfflineState	Offline State Default: 0
4 LONG	Dis_pressure_mPSI	Dispense calibrated mPSI Default: 0 Units: psi * 1000
8 LONG	Fill_pressure_mPSI	Fill calibrated PSI Default: 0 Units: psi * 1000
12 UINT16	DispensePosition	Dispense Position Default: 0 Units: ml * 1000
14 INT16	FillPosition	Fill Position Default: 0 Units: ml * 1000
16 INT16	N2Sensor_cPSI	N2 Sensor Default: 0 Units: psi * 100
18 INT16	VacSensor_cPSI	Vacuum Sensor Default: 0 Units: psi * 100

START_AUTO_REC: Start Auto_Rec

This command will execute a system auto-recirculation.

Command Code: 209 START_AUTO_REC	
Send Data	<i>None</i>
Receive Data	<i>None</i>
Access Level	OPERATOR

Status Codes

0	System Ready
1	Operation Pending
9	Start Pre-Dispense Cycle
10	Start Pre-Dispense Cycle
11	Start Pre-Dispense Cycle
12	Pre-Dispense segment 1
13	Pre-Dispense segment 2
14	Pre-Dispense segment 3
15	Pre-Dispense segment 4
16	Pre-Dispense segment 5
17	Pre-Dispense end
18	Pre-Dispense end
19	Pre-Dispense suck-back delay
20	Pre-Dispense suck-back
25	Pre-Dispense Recharge
35	Start Dispense Cycle
39	Dispense Start
40	Dispense Start
41	Dispense Start
42	Dispensing segment 1
43	Dispensing segment 2
44	Dispensing segment 3
45	Dispensing segment 4
46	Dispensing segment 5
47	Dispense end
48	Dispense end
49	Suck-back delay
50	Suck-back
55	System Recharge - START
60	System Recharge - Filling Segment
61	System Recharge - Filling Segment
62	System Recharge - Filling Segment
63	System Recharge - Filling Segment
64	System Recharge - Filling Segment
65	System Recharge - Filling Segment
66	System Recharge - Filling Segment
67	System Recharge - Filling Segment
68	System Recharge - Filling Segment
69	System Recharge - Filling Segment
70	System Recharge - Pre-filtration Segment
71	System Recharge - Pre-filtration Segment
72	System Recharge - Pre-filtration Segment

73	System Recharge - Pre-filtration Segment
74	System Recharge - Pre-filtration Segment
75	System Recharge - Pre-filtration Segment
76	System Recharge - Pre-filtration Segment
77	System Recharge - Pre-filtration Segment
78	System Recharge - Pre-filtration Segment
79	System Recharge - Pre-filtration Segment
80	System Recharge - Filtration Segment
81	System Recharge - Filtration Segment
82	System Recharge - Filtration Segment
83	System Recharge - Filtration Segment
84	System Recharge - Filtration Segment
85	System Recharge - Filtration Segment
86	System Recharge - Filtration Segment
87	System Recharge - Filtration Segment
88	System Recharge - Filtration Segment
89	System Recharge - Filtration Segment
90	System Recharge - Vent Segment
91	System Recharge - Vent Segment
92	System Recharge - Vent Segment
93	System Recharge - Vent Segment
94	System Recharge - Vent Segment
95	System Recharge - Vent Segment
96	System Recharge - Vent Segment
97	System Recharge - Vent Segment
98	System Recharge - Vent Segment
99	System Recharge - Vent Segment
100	System Recharge - Air Detection Segment
101	System Recharge - Air Detection Segment
102	System Recharge - Air Detection Segment
103	System Recharge - Air Detection Segment
104	System Recharge - Air Detection Segment
105	System Recharge - Air Detection Segment
106	System Recharge - Air Detection Segment
107	System Recharge - Air Detection Segment
108	System Recharge - Air Detection Segment
109	System Recharge - Air Detection Segment
110	System Recharge - Purge Segment
111	System Recharge - Purge Segment
112	System Recharge - Purge Segment
113	System Recharge - Purge Segment
114	System Recharge - Purge Segment
115	System Recharge - Purge Segment
116	System Recharge - Purge Segment
117	System Recharge - Purge Segment

118	System Recharge - Purge Segment
119	System Recharge - Purge Segment
120	System Recharge - Final Setup Segment
121	System Recharge - Final Setup Segment
122	System Recharge - Final Setup Segment
123	System Recharge - Final Setup Segment
124	System Recharge - Final Setup Segment
125	System Recharge - Final Setup Segment
126	System Recharge - Final Setup Segment
127	System Recharge - Final Setup Segment
128	System Recharge - Final Setup Segment
129	System Recharge - Final Setup Segment
130	System Recharge - Finish Fill Segment
140	System Recharge - In-Cycle Auto Recirculation
141	System Recharge - In-Cycle Auto Recirculation
142	System Recharge - In-Cycle Auto Recirculation
150	System Recharge - Complete
500	System Changing Home: Start
501	System Changing Home: Step 1
502	System Changing Home: Step 2
503	System Changing Home: Step 3
504	System Changing Home: Step 4
505	System Changing Home: Step 5
506	System Changing Home: Step 6
507	System Changing Home: Step 7
508	System Changing Home: Step 8
509	System Changing Home: Step 9
510	System Changing Home: Step 10
1000	System Initialization Pending
1001	Start System Initialization
1002	System Initialization
1003	System Initialization
1010	System Initialization - Fill motor init
1011	System Initialization - Fill motor init
1012	System Initialization - Fill motor init
1013	System Initialization - Fill motor init
1014	System Initialization - Fill motor init
1015	System Initialization - Fill motor init
1016	System Initialization - Fill motor init
1017	System Initialization - Fill motor init
1018	System Initialization - Fill motor init
1019	System Initialization - Fill motor init
1020	System Initialization
1021	System Initialization
1022	System Initialization

1030	System Initialization - dis motor init
1031	System Initialization - dis motor init
1032	System Initialization - dis motor init
1033	System Initialization - dis motor init
1034	System Initialization - dis motor init
1035	System Initialization - dis motor init
1036	System Initialization - dis motor init
1037	System Initialization - dis motor init
1038	System Initialization - dis motor init
1039	System Initialization - dis motor init
1050	System Initialization
1051	System Initialization
1052	System Initialization
1053	System Initialization
1054	System Initialization
1055	System Initialization
1056	System Initialization
1057	System Initialization
1058	System Initialization
1059	System Initialization
1060	System Initialization
1061	System Initialization
1070	System Vent Only Operation
1071	System Vent Only Operation
1072	System Vent Only Operation
1073	System Vent Only Operation
1074	System Vent Only Operation
1075	System Vent Only Operation
1076	System Vent Only Operation
1077	System Vent Only Operation
1078	System Vent Only Operation
1079	System Vent Only Operation
1080	System Vent Only Operation
1100	Pump Flushing - Start
1101	Pump Flushing - Filter
1102	Pump Flushing - Filter
1103	Pump Flushing - Purge
1104	Pump Flushing - Purge
1105	Pump Flushing - Vent
1106	Pump Flushing - Vent
1107	Pump Flushing
1108	Pump Flushing
1109	Pump Flushing
1110	Pump Flushing
1111	Pump Flushing

1112	Pump Flushing - End
1200	Self Test - Dispense Backlash Test
1201	Self Test - Dispense Backlash Test
1202	Self Test - Dispense Backlash Test
1203	Self Test - Dispense Backlash Test
1204	Self Test - Dispense Backlash Test
1205	Self Test - Dispense Backlash Test
1206	Self Test - Dispense Backlash Test
1207	Self Test - Dispense Backlash Test
1208	Self Test - Dispense Backlash Test
1209	Self Test - Dispense Backlash Test
1210	Self Test - Dispense Backlash Test
1211	Self Test - Dispense Backlash Test
1212	Self Test - Dispense Backlash Test
1213	Self Test - Dispense Backlash Test
1214	Self Test - Dispense Backlash Test
1215	Self Test - Dispense Backlash Test
1216	Self Test - Dispense Backlash Test
1250	Self Test - Fill Motor Strength Test
1251	Self Test - Fill Motor Strength Test
1252	Self Test - Fill Motor Strength Test
1253	Self Test - Fill Motor Strength Test
1254	Self Test - Fill Motor Strength Test
1255	Self Test - Fill Motor Strength Test
1256	Self Test - Fill Motor Strength Test
1257	Self Test - Fill Motor Strength Test
1258	Self Test - Fill Motor Strength Test
1259	Self Test - Fill Motor Strength Test
1260	Self Test - Fill Motor Strength Test
1261	Self Test - Fill Motor Strength Test
1262	Self Test - Fill Motor Strength Test
1263	Self Test - Fill Motor Strength Test
1264	Self Test - Fill Motor Strength Test
1265	Self Test - Fill Motor Strength Test
1266	Self Test - Fill Motor Strength Test
1400	Self Test - Dispense Motor Strength Test
1401	Self Test - Dispense Motor Strength Test
1402	Self Test - Dispense Motor Strength Test
1403	Self Test - Dispense Motor Strength Test
1404	Self Test - Dispense Motor Strength Test
1405	Self Test - Dispense Motor Strength Test
1406	Self Test - Dispense Motor Strength Test
1407	Self Test - Dispense Motor Strength Test
1408	Self Test - Dispense Motor Strength Test
1409	Self Test - Dispense Motor Strength Test

1410	Self Test - Dispense Motor Strength Test
1411	Self Test - Dispense Motor Strength Test
1412	Self Test - Dispense Motor Strength Test
1413	Self Test - Dispense Motor Strength Test
1414	Self Test - Dispense Motor Strength Test
1415	Self Test - Dispense Motor Strength Test
1416	Self Test - Dispense Motor Strength Test
1450	Self Test - Pressure Sensor Accuracy Test
1451	Self Test - Pressure Sensor Accuracy Test
1452	Self Test - Pressure Sensor Accuracy Test
1453	Self Test - Pressure Sensor Accuracy Test
1454	Self Test - Pressure Sensor Accuracy Test
1455	Self Test - Pressure Sensor Accuracy Test
1456	Self Test - Pressure Sensor Accuracy Test
1457	Self Test - Pressure Sensor Accuracy Test
1458	Self Test - Pressure Sensor Accuracy Test
1459	Self Test - Pressure Sensor Accuracy Test
1460	Self Test - Pressure Sensor Accuracy Test
1461	Self Test - Pressure Sensor Accuracy Test
1462	Self Test - Pressure Sensor Accuracy Test
1463	Self Test - Pressure Sensor Accuracy Test
1464	Self Test - Pressure Sensor Accuracy Test
1465	Self Test - Pressure Sensor Accuracy Test
1466	Self Test - Pressure Sensor Accuracy Test
1475	Self Test - Pressure Re-Zero Test
1476	Self Test - Pressure Re-Zero Test
1477	Self Test - Pressure Re-Zero Test
1478	Self Test - Pressure Re-Zero Test
1479	Self Test - Pressure Re-Zero Test
1480	Self Test - Pressure Re-Zero Test
1481	Self Test - Pressure Re-Zero Test
1482	Self Test - Pressure Re-Zero Test
1483	Self Test - Pressure Re-Zero Test
1484	Self Test - Pressure Re-Zero Test
1485	Self Test - Pressure Re-Zero Test
1486	Self Test - Pressure Re-Zero Test
1487	Self Test - Pressure Re-Zero Test
1488	Self Test - Pressure Re-Zero Test
1489	Self Test - Pressure Re-Zero Test
1490	Self Test - Pressure Re-Zero Test
1491	Self Test - Pressure Re-Zero Test
1500	Self Test - Pump Leak Test
1501	Self Test - Pump Leak Test
1502	Self Test - Pump Leak Test
1503	Self Test - Pump Leak Test

1504	Self Test - Pump Leak Test
1505	Self Test - Pump Leak Test
1506	Self Test - Pump Leak Test
1507	Self Test - Pump Leak Test
1508	Self Test - Pump Leak Test
1509	Self Test - Pump Leak Test
1510	Self Test - Pump Leak Test
1511	Self Test - Pump Leak Test
1512	Self Test - Pump Leak Test
1513	Self Test - Pump Leak Test
1514	Self Test - Pump Leak Test
1515	Self Test - Pump Leak Test
1516	Self Test - Pump Leak Test
1530	Self Test - Air Detect
1531	Self Test - Air Detect
1532	Self Test - Air Detect
1533	Self Test - Air Detect
1534	Self Test - Air Detect
1535	Self Test - Air Detect
1536	Self Test - Air Detect
1537	Self Test - Air Detect
1538	Self Test - Air Detect
1539	Self Test - Air Detect
1540	Self Test - Air Detect
1541	Self Test - Air Detect
1542	Self Test - Air Detect
1543	Self Test - Air Detect
1544	Self Test - Air Detect
1545	Self Test - Air Detect
1546	Self Test - Air Detect
1600	Self Test - Pump Valve Function Test
1601	Self Test - Pump Valve Function Test
1602	Self Test - Pump Valve Function Test
1603	Self Test - Pump Valve Function Test
1604	Self Test - Pump Valve Function Test
1605	Self Test - Pump Valve Function Test
1606	Self Test - Pump Valve Function Test
1607	Self Test - Pump Valve Function Test
1608	Self Test - Pump Valve Function Test
1609	Self Test - Pump Valve Function Test
1610	Self Test - Pump Valve Function Test
1611	Self Test - Pump Valve Function Test
1612	Self Test - Pump Valve Function Test
1613	Self Test - Pump Valve Function Test
1614	Self Test - Pump Valve Function Test

1615	Self Test - Pump Valve Function Test
1616	Self Test - Pump Valve Function Test
1700	Self Test - High Viscosity Valve Timing Test
1701	Self Test - High Viscosity Valve Timing Test 1
1702	Self Test - High Viscosity Valve Timing Test 2
1703	Self Test - High Viscosity Valve Timing Test 3
1704	Self Test - High Viscosity Valve Timing Test 4
1705	Self Test - High Viscosity Valve Timing Test 5
1706	Self Test - High Viscosity Valve Timing Test 6
1707	Self Test - High Viscosity Valve Timing Test 7
1708	Self Test - High Viscosity Valve Timing Test 8
1709	Self Test - High Viscosity Valve Timing Test 9
1710	Self Test - High Viscosity Valve Timing Test 10
1711	Self Test - High Viscosity Valve Timing Test
1712	Self Test - High Viscosity Valve Timing Test
1713	Self Test - High Viscosity Valve Timing Test
1714	Self Test - High Viscosity Valve Timing Test
1715	Self Test - High Viscosity Valve Timing Test
1716	Self Test - High Viscosity Valve Timing Test Complete
2000	Auto Rec Start
2001	Auto Rec step 1
2002	Auto Rec step 2
2003	Auto Rec step 3
2004	Auto Rec step 4
2005	Auto Rec step 5
3000	Start Cycle Internal Valves
3001	Stop Cycle Internal Valves
4000	Strobe recipe is active
4500	Pressure balance is active
4990	Clear Error Pending
4995	Downloading Firmware
5000	Reference Profile Saving
5001	Reference Profile Saving - Recipe 1
5002	Reference Profile Saving - Recipe 2
5003	Reference Profile Saving - Recipe 3
5004	Reference Profile Saving - Recipe 4
5005	Reference Profile Saving - Recipe 5
5006	Reference Profile Saving - Recipe 6
5007	Reference Profile Saving - Recipe 7
5008	Reference Profile Saving - Recipe 8
5009	Reference Profile Saving - Recipe 9
5010	Reference Profile Saving - Recipe 10
5011	Reference Profile Saving - Recipe 11
5012	Reference Profile Saving - Recipe 12
5100	Reference Profile Deleting

5101	Reference Profile Deleting - Recipe 1
5102	Reference Profile Deleting - Recipe 2
5103	Reference Profile Deleting - Recipe 3
5104	Reference Profile Deleting - Recipe 4
5105	Reference Profile Deleting - Recipe 5
5106	Reference Profile Deleting - Recipe 6
5107	Reference Profile Deleting - Recipe 7
5108	Reference Profile Deleting - Recipe 8
5109	Reference Profile Deleting - Recipe 9
5110	Reference Profile Deleting - Recipe 10
5111	Reference Profile Deleting - Recipe 11
5112	Reference Profile Deleting - Recipe 12
7777	System Ready - OFFLINE
8887	System is in Maintenance Mode with a Timeout
8888	System is in Maintenance Mode
9998	System Error
9999	Critical System Error
10001	System is not connected
10002	System attempting to connect
20000	System Priming: Start Priming
20001	System Priming: Priming Step 1
20002	System Priming: Priming Step 2
20003	System Priming: Priming Step 3
20004	System Priming: Priming Step 4
20005	System Priming: Priming Step 5
20006	System Priming: Priming Step 6
20007	System Priming: Priming Step 7
20008	System Priming: Priming Step 8
20009	System Priming: Priming Step 9
20010	System Priming: Priming Step 10
20011	System Priming: Priming Step 11
20012	System Priming: Priming Step 12
20013	System Priming: Priming Step 13
20014	System Priming: Priming Step 14
20015	System Priming: Priming Step 15
20016	System Priming: Priming Step 16
20017	System Priming: Priming Step 17
20018	System Priming: Priming Step 18
20019	System Priming: Priming Step 19
20020	System Priming: Priming Step 20
20021	System Priming: Priming Step 21
20022	System Priming: Priming Step 22
20023	System Priming: Priming Step 23
20024	System Priming: Priming Step 24
20025	System Priming: Priming Step 25

20026	System Priming: Priming Step 26
20027	System Priming: Priming Step 27
20028	System Priming: Priming Step 28
20029	System Priming: Priming Step 29
20030	System Priming: Priming Step 30
20031	System Priming: Finish Priming
20032	System Priming: Priming Paused

Return Codes

0	Good
1	Size Error
2	Unknown Command
3	Bad CRC
4	Comm Error
5	Non volatile Error
6	The system is busy with a previous command. Please wait for READY status, then try again.
7	Unknown Internal Error (should not happen)
8	Bad Diagnostic
9	Diag failed
10	One or more of the values you just entered is above the maximum value for that parameter. The problem fields are highlighted in red.
11	One or more of the values you just entered is below the minimum value for that parameter. The problem fields are highlighted in red.
12	SVID List too large
13	Unknown SVID
14	Flash too many segments
15	Flash Illegal Address
16	Flash segment overlap
17	Flash no segment
18	Flash Duplicate
19	Download not sequential
20	Error in firmware image (number of segments)
21	Error in firmware image (CRC)
22	No firmware validation code received
23	Dataflash failure
24	Dataflash verify failure
25	Dataflash failed CRC
26	Dataflash read size mismatch
27	No records available
28	Error decoding trace data
29	No profile for this cycle
30	Bounds error for profile data start
31	Serial Trigger commands have been disabled
32	Bad password
33	Dataflash chiperror failure
34	System is already online
35	System is already offline
36	Command not supported
37	The system is currently offline.
38	The system requires a power cycle or software reset.
70	Bad Recipe Number

71	Bad sequence number
72	Bad recipe number
73	Max dispense volume + purge volume is too high
74	The first step cannot be a loop
76	Total recipe volume too low
77	Total recipe volume too high
78	Dispense segment 1 is incomplete
79	Dispense segment 2 is incomplete
80	Dispense segment 3 is incomplete
81	Dispense segment 4 is incomplete
82	Dispense segment 5 is incomplete
83	This interface module does not have flow sensor capability. Please make sure the flow sensor is disabled in the system parameters.
84	The current system configuration has no control over the outlet valve
85	Invalid "Go to Step" value for the Loop step-type
86	One or more dispense segments are incomplete
255	This is the max code

Trace Bits

1	Dispense Pressure
2	Fill Pressure
4	Dispense Position
8	Fill Position
16	Dispense Current
32	System Status
64	Valves Values
128	Flow Sensor
256	N2 Supply
512	Vacuum Supply

Trace Numbers

0	Dispense Pressure
1	Fill Pressure
2	Dispense Position
3	Fill Position
4	Dispense Current
5	System Status
6	Valves Values
7	Flow Sensor
8	N2 Supply
9	Vacuum Supply

Product IDs

0	IntelliGen LV	
	Dispense System	Low viscosity pump without Anti-backlash
1	IntelliGen MV	
	Dispense System	Mid viscosity/pressure pump
2	IntelliGen ULV	
	Dispense System	Ultra low viscosity/pressure pump with Anti-backlash
3	AFS10 10mL	Ultra low viscosity/pressure pump with Anti-backlash, XSV-10ml version
4	IntelliGen XSV	
	Dispense System	Low viscosity pump without Anti-backlash, XSV-3ml version
5	IntelliGen RULV	
	Dispense System	Ultra low viscosity/pressure pump with Anti-backlash in RDS01 format
6	IntelliGen HV20	
	Dispense System	High viscosity/pressure pump 20 cc version
7	IntelliGen HV50	
	Dispense System	High viscosity/pressure pump 50 cc version
8	IntelliGen HMV	
	Dispense System	Mid-High viscosity/pressure pump 20 cc version

SVIDs

SVID	Size	Type	Data	Description
1	24	STRING	SerialNumbers.Pump[24]	Pump Serial Number
2	24	STRING	SerialNumbers.FillMotor[24]	Fill Motor Serial Number
3	24	STRING	SerialNumbers .DispenseMotor[24]	Dispense Motor Serial Number
4	24	STRING	SerialNumbers.FillPressure[24]	Fill Pressure Serial Number
5	24	STRING	SerialNumbers .DispensePressure[24]	Dispense Pressure Serial Number
6	24	STRING	SerialNumbers.MainBoard[24]	Main PCB Serial Number
7	24	STRING	SerialNumbers .InternalBoard[24]	Internal PCB Serial Number
8	24	STRING	SerialNumbers .InterfaceBoard[24]	Interface PCB Serial Number
10	24	STRING	Version.Model[24]	Model Name
11	48	STRING	Version.Version[48]	Firmware Version String
12	24	STRING	SystemNames.CustomerName[24]	Customer Name
13	24	STRING	SystemNames.FabName[24]	Fab Name
14	24	STRING	SystemNames.ToolName[24]	Tool Name
15	24	STRING	SystemNames.CoaterName[24]	Coater Name
16	24	STRING	SystemNames.NozzleName[24]	Nozzle Name
17	24	STRING	SystemNames.ResistName[24]	Resist Name
18	24	STRING	SystemNames.PumpName[24]	Pump Name
19	24	STRING	SystemNames.FluidName[24]	Fluid Name/Type
20	24	STRING	Recipe.RecipeTitle[24]	Description
21	24	STRING	DispenseInfo.JobInfo[24]	Job Information
257	2	INT16	RtStatus.Status	System Status
258	2	INT16	UtilityStatus.HomingCurrent_mA	Peak homing current of DC motor
259	2	INT16	UtilityStatus.reserved1	reserved1
260	2	INT16	UtilityStatus .DisCurrent_UnCal_mA	Dispense Motor Current
261	2	INT16	UtilityStatus .DispenseCurrent_mA	Dispense Motor Current
263	1	INT16	FillMotor.Position_uL	Fill Position in uL
264	2	INT16	Recharge.AutoVentEnable	Auto-Vent Enable
265	2	INT16	Recharge.VentFreqEnable	Vent Frequency Enable

266	2	INT16	Recipe.OpenAdjust	Outlet Valve Open Adj.
267	2	INT16	Recipe.CloseAdjust	Outlet Valve Close Adj.
268	2	INT16	Recipe.VolumeCalibration	Volume Calibration
269	2	INT16	PressureStatus .Dis_pressure_cPSI	dispense calibrated cPSI
270	2	INT16	PressureStatus .Fill_pressure_cPSI	Fill calibrated cPSI
275	2	INT16	DispenseInfo.ReadyPressure	Ready - Ready Pressure
276	2	INT16	DispenseInfo.PeakPressure	Dispense - Maximum Pressure
277	2	INT16	DispenseInfo .ReferencePeakPressure	Reference Dispense - Maximum Pressure
278	2	INT16	DispenseInfo.AveragePressure	Dispense - Average Pressure
279	2	INT16	DispenseInfo .ReferenceAveragePressure	Reference Dispense - Average Pressure
280	2	INT16	DispenseInfo.CutoffPressure	Dispense - Cutoff Pressure
281	2	INT16	DispenseInfo .ReferenceCutoffPressure	Reference Dispense - Cutoff Pressure
282	2	INT16	DispenseInfo.MaxFiltrationPrs	Filtration - Maximum Upstream Pressure
283	2	INT16	DispenseInfo .ReferenceMaxFiltrationPrs	Reference Filtration - Maximum Upstream Pressure
284	2	INT16	DispenseInfo.MinFillPrs	Fill - Minimum Fill Pressure
285	2	INT16	DispenseInfo .ReferenceMinFillPrs	Reference Fill - Minimum Fill Pressure
286	2	INT16	SkipstepInfo.StepMismatch_uL	StepMismatch uL
287	2	INT16	UtilityStatus.N2Sensor_cPSI	N2 Sensor
288	2	INT16	UtilityStatus.VacSensor_cPSI	Vacuum Sensor
289	2	INT16	DispenseInfo .ReferenceReadyPressure	Reference Ready - Ready Pressure
290	2	INT16	DispenseInfo.AverageCurrent	Dispense - Average Motor Current
291	2	INT16	DispenseInfo .ReferenceAverageCurrent	Reference Dispense - Average Motor Current
292	2	INT16	DispenseInfo.PredictedCurrent	Dispense - Predicted Motor Current
293	2	INT16	DispenseInfo .ReferencePredictedCurrent	Reference Dispense - Predicted Motor Current
294	2	INT16	DispenseInfo .AverageDeltaPressure	Filtration - Average Delta Pressure
295	2	INT16	DispenseInfo .ReferenceAverageDeltaPressure	Reference Filtration - Average Delta Pressure
296	2	INT16	DispenseInfo.FillMotorMismatch	Cycle - Fill Motor Mismatch

297	2	INT16	DispenseInfo .ReferenceFillMotorMismatch	Cycle Dispense - Fill Motor Mismatch
298	2	INT16	UtilityStatus .TempFillSensor_cC	Fill Temperature
299	2	INT16	UtilityStatus.TempDisSensor_cC	Dispense Temperature
300	2	INT16	UtilityStatus.N2SensorTemp_cC	N2 Sensor Temp
301	2	INT16	UtilityStatus.VacSensorTemp_cC	Vacuum Sensor Temp
302	2	INT16	RtStatus.OfflineState	OfflineState
513	4	FLOAT	PressureStatus .Dis_pressure_PSI	dispense calibrated PSI
514	4	FLOAT	PressureStatus .Fill_pressure_PSI	Fill calibrated PSI
769	2	UINT16	FirmwareInfo.ProductQualifier	Qualifer for ProductCode to be valid
770	2	UINT16	FirmwareInfo.Comms_Version	Communications protocol version
771	2	UINT16	FirmwareInfo.MajorVersion	Major Version Number
772	2	UINT16	FirmwareInfo.ModuleType	Interface Module Type
773	2	UINT16	FirmwareInfo.MinorVersion	Firmware Minor Version
774	2	UINT16	FirmwareInfo.ProductCode	Product ID Code
775	2	UINT16	SystemParameters.Address	Serial Address
776	2	UINT16	SystemParameters .FluidViscosity_cP	Fluid Viscosity
778	2	UINT16	SystemParameters .ReadyPressureEnable	Ready Pressure Enable
780	2	UINT16	SystemParameters.AutoRecEnable	Auto-Recirculation
781	2	UINT16	SystemParameters .AutoRecVolume_uL	Auto-Recirc. Volume
782	2	UINT16	SystemParameters .AutoRecRate_uLS	Auto-Recirc. Rate
783	2	UINT16	SystemParameters .MaintenanceModeTimeoutEnable	Maint. Mode Timeout
784	2	UINT16	SystemParameters .MaintenanceModeTimeout_min	Maint. Mode Timeout
795	2	UINT16	RtStatus.LastRecipeNum	Last Dispensed Recipe Num
796	2	UINT16	RtStatus.LastDispenseVol_uL	Last Dispensed Total volume
797	2	UINT16	RtStatus .LastDispenseConfirmation_per	Last Dispensed confirmation value
799	2	UINT16	RtStatus.CriticalErrors	counts Critical System Errors after they are logged
800	2	UINT16	RtStatus.Errors	counts System Errors after they are logged

801	2	UINT16	RtStatus.Warnings	counts System Warnings after they are logged
802	2	UINT16	RtStatus.Infos	counts System infos after they are logged
803	2	UINT16	RtStatus.TotalAlarms	counts total uncleared alarms after they are logged
808	1	UINT16	DisMotor.Position_uL	Dispense Position in uL
809	2	UINT16	SystemParameters .SerialTriggerEnable	Serial Trigger Enable
810	2	UINT16	SystemParameters.reserved8	reserved
811	2	UINT16	SystemParameters .VentOnlyVolume_uL	Vent Only Volume
812	2	UINT16	SystemParameters .VentOnlyRate_uLS	Vent Only Rate
813	2	UINT16	SystemParameters .PreFiltrationAlarmLevel	Pre-Filtration Alarm Level
814	2	UINT16	SystemParameters .FiltrationAlarmLevel	Filtration Alarm Level
815	2	UINT16	SystemParameters .UtilityAlarmLevel	Utility Alarm Level
816	2	UINT16	Recharge.VentFreq	Vent Frequency
817	2	UINT16	Recharge.FreqVentVolume	Vent Volume (Frequency)
818	2	UINT16	Recharge.FreqVentRate	Vent Rate (Frequency)
819	2	UINT16	Recharge.PurgeVolume_uL	Purge Volume
820	2	UINT16	Recharge.PurgeRate	Purge Rate
821	2	UINT16	Recharge.FillRate	Fill Rate
822	2	UINT16	Recharge.FiltrationRate	Filtration Rate
823	2	UINT16	Recharge .VariableDispenseVolume	Max. Dispense Volume
824	2	UINT16	Recipe.RecipeNum	Recipe number
825	2	UINT16	Recipe.DisRampUpTime	Ramp Up Time
826	2	UINT16	Recipe.DispenseVolume_1	Segment 1 Volume
827	2	UINT16	Recipe.DispenseRate_1	Segment 1 Rate
828	2	UINT16	Recipe.DispenseVolume_2	Segment 2 Volume
829	2	UINT16	Recipe.DispenseRate_2	Segment 2 Rate
830	2	UINT16	Recipe.DispenseVolume_3	Segment 3 Volume
831	2	UINT16	Recipe.DispenseRate_3	Segment 3 Rate
832	2	UINT16	Recipe.DispenseVolume_4	Segment 4 Volume
833	2	UINT16	Recipe.DispenseRate_4	Segment 4 Rate

834	2	UINT16	Recipe.DispenseVolume_5	Segment 5 Volume
835	2	UINT16	Recipe.DispenseRate_5	Segment 5 Rate
836	2	UINT16	Recipe.DisRampDownTime	Ramp Down Time
837	2	UINT16	Recipe.PreDispenseRecipe	Pre-Dispense Recipe
838	2	UINT16	DispenseInfo .ReferenceCycleTime	Reference Cycle - Total Time
839	2	UINT16	DispenseInfo.CycleTime	Cycle - Total Time
840	2	UINT16	DispenseInfo.FillTime	Fill - Total Time
847	2	UINT16	DispenseInfo.RecipeNum	Recipe number
848	2	UINT16	DispenseInfo.SlotID	Slot ID
849	2	UINT16	DispenseInfo.Confirmation	Dispense - Pressure Profile Compare
850	2	UINT16	DispenseInfo .ReferenceConfirmation	Reference Dispense - Pressure Profile Compare
851	2	UINT16	DispenseInfo.AverageRate	Dispense - Average Motor Rate
852	2	UINT16	DispenseInfo .ReferenceAverageRate	Reference Dispense - Average Motor Rate
853	2	UINT16	DispenseInfo.TotalVolume	Dispense - Total Motor Volume
854	2	UINT16	DispenseInfo .ReferenceTotalVolume	Reference Dispense - Total Motor Volume
855	2	UINT16	DispenseInfo.AirDetect	Dispense - Air Detect Volume
856	2	UINT16	DispenseInfo .ReferenceAirDetect	Reference Dispense - Air Detect Volume
858	2	UINT16	DispenseInfo.FlowConfirmation	Dispense - Flow Profile Compare
859	2	UINT16	DispenseInfo .ReferenceFlowConfirmation	Reference Dispense - Flow Profile Compare
860	2	UINT16	DispenseInfo.ReferenceFillTime	Reference Fill - Total Time
861	2	UINT16	DispenseInfo.FiltrationRate	Filtration - Average Motor Rate
862	2	UINT16	DispenseInfo .ReferenceFiltrationRate	Reference Filtration - Average Motor Rate
863	2	UINT16	DispenseInfo.FiltrationTime	Filtration - Total Time
864	2	UINT16	DispenseInfo .ReferenceFiltrationTime	Reference Filtration - Total Time
866	2	UINT16	DispenseInfo.AverageFillRate	Fill - Average Motor Rate
867	2	UINT16	DispenseInfo .ReferenceAverageFillRate	Reference Fill - Average Motor Rate
868	2	UINT16	DispenseInfo.VentTime	Vent - Total Time
869	2	UINT16	DispenseInfo.ReferenceVentTime	Reference Vent - Total Time

870	2	UINT16	DispenseInfo .ConfirmationMiniStyle	Dispense - Pres Profile Cmp (IGMINI style)
871	2	UINT16	DispenseInfo .ReferenceConfirmationMiniStyle	Reference Dispense - Pres Profile Cmp (IGMINI style)
1025	4	ULONG	TimeNow.Time	UTC seconds from 1 Jan 1970
1026	4	ULONG	CycleCounters.CycleCounts	Total Cycles
1027	4	ULONG	CycleCounters.PowerCycles	Total Power Cycles
1028	4	ULONG	CycleCounters.DispenseVolume	Total Dispensed Volume (L)
1029	4	ULONG	CycleCounters.FiltratedVolume	Total Filtrated Volume (L)
1030	4	ULONG	CycleCounters.VentedVolume	Total Vented Volume (L)
1031	4	ULONG	CycleCounters.PurgedVolume	Total Purged Volume (L)
1032	4	ULONG	CycleCounters.FilledVolume	Total Filled Volume (L)
1033	4	ULONG	CycleCounters .CycleCounts_reset_A	Resettable A Cycles
1034	4	ULONG	CycleCounters .PowerCycles_reset_A	Resettable A Power Cycles
1035	4	ULONG	CycleCounters .DispenseVolume_reset_A	Resettable A Dispensed Volume (L)
1036	4	ULONG	CycleCounters .FiltratedVolume_reset_A	Resettable A Filtrated Volume (L)
1037	4	ULONG	CycleCounters .VentedVolume_reset_A	Resettable A Vented Volume (L)
1038	4	ULONG	CycleCounters .PurgedVolume_reset_A	Resettable A Purged Volume (L)
1039	4	ULONG	CycleCounters .FilledVolume_reset_A	Resettable A Filled Volume (L)
1040	4	ULONG	CycleCounters .CycleCounts_reset_B	Resettable B Cycles
1041	4	ULONG	CycleCounters .PowerCycles_reset_B	Resettable B Power Cycles
1042	4	ULONG	CycleCounters .DispenseVolume_reset_B	Resettable B Dispensed Volume (L)
1043	4	ULONG	CycleCounters .FiltratedVolume_reset_B	Resettable B Filtrated Volume (L)
1044	4	ULONG	CycleCounters .VentedVolume_reset_B	Resettable B Vented Volume (L)
1045	4	ULONG	CycleCounters .PurgedVolume_reset_B	Resettable B Purged Volume (L)
1046	4	ULONG	CycleCounters .FilledVolume_reset_B	Resettable B Filled Volume (L)

1047	4	ULONG	SystemParameters .FilterCounterLimit	Filter Cycle Alarm Limit
1048	4	ULONG	RtStatus.NewestAlarmNum	Number of newest alarm
1049	4	ULONG	RtStatus.NewestRecordNum	Number of newest record
1056	4	ULONG	RtStatus.LastDispenseCycle	Last Dispensed Cycle
1057	4	ULONG	RtStatus.LastClearedRecord	# of the last cleared record
1058	4	ULONG	RtStatus.ProfileCycle	Most recent profile cycle
1059	4	ULONG	SystemParameters.Time	Last saved
1060	4	ULONG	SystemParameters.Cycle	Cycle when written
1061	4	ULONG	Recharge.Time	Last saved
1062	4	ULONG	Recharge.Cycle	Cycle when written
1063	4	ULONG	Recipe.Time	Last saved
1064	4	ULONG	Recipe.Cycle	Cycle when written
1069	4	ULONG	DispenseInfo.Cycle	Cycle Number
1070	4	ULONG	DispenseInfo.ReferenceCycle	Reference Cycle Number
1071	4	ULONG	DispenseInfo.Time	Time of dispense (trigger)
1072	4	ULONG	SkipstepInfo.Cycle	Cycle for this information
1285	4	LONG	SystemParameters .ReadyPressureSetpoint_mPSI	RDY Pressure Setpoint
1286	4	LONG	SystemParameters .AutoRecTimer_min	Auto-Recirc. Timer
1293	4	LONG	PressureStatus .Dis_pressure_mPSI	dispense calibrated mPSI
1294	4	LONG	PressureStatus .Fill_pressure_mPSI	Fill calibrated mPSI
1295	4	LONG	Recharge .FiltrationPressureSetpoint	Filtration psi Setpoint

Appendix

Alarms

Alarms are informational records used to log events that occur in the pump. Information about individual alarm codes are in a separate alarms document. Each alarm has a code, a timestamp, a unique ID number, and 4 bytes of data. The ID number is a unique number for each alarm, which is used for alarm retrieval. Alarms are classified as one of 5 types:

None: 0 - Not classified

Information: 1 - Information only

Warning: 2 - Initiates Warning condition in the pump

Error: 3 - Initiates Error condition in the pump

Critical: 4 - Disables the motors and initiates Error condition in the pump

Alarms are retrieved with the GET_ALARMS command by sending the first alarm number to retrieve and a count of how many alarms to return. As a special case alarm number 0 for Start is always the most recent alarm.

NOTE: The GET_ALARMS command may have to search through thousands of log records to find alarm log records. To eliminate this possibly long time and stop communication timeouts the GET_ALARMS command will terminate after 500mS of searching and gathering and return without all of the requested alarms. It may take several subsequent commands to get all of the desired alarms. The most recent alarm number can also be found as TotalAlarms in the RTSTATUS command.

The data for each alarm is:

UINT16 Size 14 words (28 bytes)

UINT16 RecordType 1 for alarm record

ULONG CycleNum "Pump cycle number"

ULONG TimeStamp "date and time stamp"

ULONG RecordNumber

ULONG AlarmNumber "Alarm Number"

UINT16 Code "Alarm Code -- defined below"

UINT16 AlarmSeverity "Info, warning, error, critical error"

ULONG Data "Alarm specific data"

Alarm specific data can be decoded and used in the strings as variables below.

The long (4 byte) data is decoded like this:

BYTE0 BYTE1 BYTE2 BYTE3

WORD0 WORD1

LONG

The variables used are shown here.

\$ULONG LONG Display as integer

\$LONG LONG (signed value) Display as integer
\$SHORT0 WORD0 (signed value) Display as integer
\$SHORT1 WORD1 (signed value) Display as integer
\$PRESSURE0 WORD0 centi-PSI (signed value) Display as pressure value in MMI pressure units (PSI/Bar)
\$PRESSURE1 WORD1 centi-PSI (signed value) Display as pressure value in MMI pressure units (PSI/Bar)
\$ML_0 WORD0 Display as mL divide by 1000
\$ML_1 WORD1 Display as mL divide by 1000
\$MLS_0 WORD0 Display as mL/sec divide by 1000
\$MLS_1 WORD1 Display as mL/sec divide by 1000
\$USHORT0 WORD0 Display as integer
\$USHORT1 WORD1 Display as integer
\$CSECS0 WORD0 .01 of a second Display as float: 1.23
\$CSECS1 WORD1 .01 of a second Display as float: 1.23
\$TIME LONG UTC time since 1970, display as string using local TZ
\$MODULE LONG Display as a string description of the module type
\$CPERC0 WORD0 Display as % divide by 100
\$CPERC1 WORD1 Display as % divide by 100

Bounds Check Errors

When a write command generates bounds limit errors the return code from the command is the first bounds error detected. The command will also return a list of error codes and field index for all bounds errors detected. This way the MMI or track software can display all of the errors at once.

For each bounds limit error 2 UINT16 values are returned in the data as the error code and the field index. Field indices start at 0 for the first field, 1 for the second, etc.

Passwords

The password key can be used to restrict access to certain pump features. The access levels are:

NONE: No password key is required. All pump operations can be accessed.

OPERATOR: An operator password key is required to dispense the pump, perform priming and clear errors. This is for routine operational functions of the pump but does not allow any changes to system, recharge or recipe settings.

ENGINEER: An engineer password key is required to download firmware, change system settings, recipes, etc. Engineer access includes Operator level access.

OWNER: An owner password is required to set password keys. Owner access has unlimited access to the system.

By default, password keys are disabled. Once enabled, the pump requires an additional USHORT key value of data containing a key. The pump will check each command for the required level of access and if the check fails, a bad password key (32) is returned.

Keys can be any value and are not encrypted. They are intended to be a hashed value of a user-friendly text string. The pump does not see the text strings making it agnostic to the hash algorithm. The MMI uses the same CRC16 algorithm to create the key as is used for serial data integrity.

The password command is write-only. If passwords are disabled then any key value for Owner is accepted. If passwords are enabled then the correct Owner key must be provided in the Owner field of the command for the new values to be accepted. If the Owner key is lost then the CRC16 hash of the word 'Entegris' (45021) will be accepted only for the password command to disable the passwords (and then set a new owner key).

Priming

A priming sequence can be started manually by sending a START_PRIMING command (command code 76). That command is sent with a 2 byte value representing the priming sequence to run or the priming operation to perform. Valid priming sequences are 1 - 20.

START_PRIMING command values:

- 2 - Pauses/Resumes the priming operation
- 1 - Stops the priming operation
- 0 - No action/reserved for future use
- 1 - Starts priming sequence 1
- 2 - Starts priming sequence 2
- 3 - Starts priming sequence 3
- 4 - Starts priming sequence 4
- etc...
- 20 - Starts priming sequence 20

Priming Types definitions:

- 0 - PRIMNG_CYCLE_TYPE_NONE
- 1 - PRIMNG_CYCLE_TYPE_LOOP
- 2 - PRIMNG_CYCLE_TYPE_VENT
- 3 - PRIMNG_CYCLE_TYPE_OUTLET
- 4 - PRIMNG_CYCLE_TYPE_PURGE
- 5 - PRIMNG_CYCLE_TYPE_INLET
- 6 - PRIMNG_CYCLE_TYPE_SOAK
- 7 - PRIMNG_CYCLE_TYPE_BACKFLUSH_VENT
- 8 - PRIMNG_CYCLE_TYPE_BACKFLUSH_INLET
- 9 - PRIMNG_CYCLE_TYPE_FILTER_FLUSH
- 10 - PRIMNG_CYCLE_TYPE_DIS_TO_VENT
- 11 - PRIMNG_CYCLE_TYPE_FLUSH_BUBBLES

The pump will calculate the estimated time and fluid consumption of a sequence when the sequence is sent to the pump to be saved. When reading back a sequence that was saved the estimated time and fluid consumption will be displayed in the fields below:

- 908: ULONG EstTime Est Total time to complete Default: 0
- 912: ULONG EstOutletVol Est total volume out the outlet Default: 0
- 916: ULONG EstVentVol Est total volume out the vent Default: 0
- 920: ULONG EstFilteredVol Est total volume through the filter Default: 0

The system can also validate a sequence and calculate these values "on the fly" by writing the sequence to the scratch sequence (0). If the sequence is correct the command will calculate the information and display it in the fields for sequence (0).

Profiles

The dispense system temporarily records a high resolution (5mS sample) profile of all of the trace sources. Each profile is collected separately (no interleaving of the data). Profiles are identified by cycle number and will persist for 3 cycles or until the pump is reset (powered off). Additionally, the saved reference profiles are stored in non volatile memory and will always be available until replaced with a new reference.

To get the profile data first send the GET_PROFILE_TRACE_INFO command with the the cycle number to select which dispense cycle to retrieve The data returned includes:

Encoding: This indicates if the trace data is encoded or not (0). Currently profile trace data is encoded with custom Entegris compression described below(1).

Time: Timestamp of the start of the dispense (trigger).

StartCycle: This is the sample number in the data for the trigger (the pump will record some time before the trigger).

DispenseEnd: Sample number for end of the dispense.

EndFiltration: Sample number for end of filtration.

EndCycle: Sample number for end of the cycle.

FillEnd: Sample number for end of the dispense.

UnencodedSize: The total number of samples (includes time after EndCycle).

EncodedSize[8]: Compressed size for each trace, (until compression is implemented this will always be the same UnencodedSize). Traces are indexed with Trace Numbers (below).

Trace Numbers

- 0 - Dispense Pressure
- 1 - Fill Pressure
- 2 - Dispense Position
- 3 - Fill Position
- 4 - Dispense Current
- 5 - System Status
- 6 - Valves Values
- 7 - Flow Sensor

To get the data trace data send the GET_PROFILE_TRACE command with the Cycle, Trace Number. Since traces can contain much more data than can fit in one command the data has to be read with multiple commands. To get a segment of data send the Size request and the Start. Size must not be greater than 500 samples (16 bit samples).

Profile Compression

To reduce internal memory requirements and allow more data to be uploaded faster, the dispense system uses a proprietary algorithm to compress individual profile traces. The algorithm examines the data and encodes it into pieces using one of three methods: run length coding (RLC), delta coding or uncompressed (non-encoded). This method is designed to provide the best compression for the different kinds of data while still being simple. For pressure traces and motor positions it will typically compress to 1/3 of the original size using delta encoding. For IO bits and motor positions when the

motor is not moving the entire trace of thousands of samples can be reduced to a few dozen words.

Data is encoded and decoded as a string of 16 bit words (unsigned short). The following C code is provided as-is for reference. Contact Entegris for examples in Python or C#.

```
// like memset but for 16 bit words (unsigned short on most compilers)
//
void wordset(unsigned short* wp, unsigned short w, unsigned n)
{
    while(n--) {
        *wp+++=w;
    }
}

long decode(unsigned short *e, unsigned short *u, long e_size, long max_u_size)
{
    unsigned long u_size = 0;
    unsigned short c;
    short n;
    int i;
    int delta;
    unsigned short dd;

    while (e_size > 0) {
        n = (short) *e++;
        e_size--;

        // is it a run?
        //
        if (n < 0) {
            n = -n;
            c = *e++;
            e_size--;

            // Don't over run the bffer
            //
            if ((n+u_size) > max_u_size) {
                break;
            }

            wordset(u,c,n);
            u_size += n;

            u += n;
        }
        else {

            // is it a delta?
```

```
//
if (n >= 0x4000) {
    n -= 0x4000;

    c = *e++;
    e_size--;

    // get the starting value from which the deltas originate
    //

    // Don't over run the buffer
    //
    if (u_size >= max_u_size) {
        break;
    }

    *u++=c;
    u_size++;

    while (n > 0) {
        dd = *e++;
        e_size--;
        n -= 3;
        for (i=0; i<3; i++) {
            delta = dd >> 10;
            if (delta > 15) {
                delta = delta - 32;
            }

            dd = (dd << 5) & 0x7fff;
            c = c - delta;
            //    c = c & 0xffff;

            // Don't over run the buffer
            //
            if (u_size >= max_u_size) {
                break;
            }

            *u++=c;
            u_size++;
        }
    }
}
else {

    // Don't over run the buffer
    //
    if ((n*sizeof(unsigned short)+u_size) > max_u_size) {
        break;
    }
}
```

```
    // N uncompressed words
    //
    memcpy(u,e,n*sizeof(unsigned short));
    e_size -= n*sizeof(unsigned short);
    u_size += n*sizeof(unsigned short);
    u += n;
    e += n;
  }
}
}

return u_size;
}
```

Real Time Traces

The real time traces will return recent real time data for many samples and traces in one command. The command was created to allow efficient transfer of about 10 samples per second and to allow this data to be aligned with no jitter introduced by the serial or network communications to the pump.

Typically, the pump is polled with this command every second for 12-15 points at a time. The first ULONG (4 bytes) is the latest sample index and it used to calculate the next data point to display.

The real time data is collected internally every 100mS for all the traces. This command can provide more than 10 of the most recent samples. By getting more than 10 (1 second) of data any gaps caused by delays or retries of the command can be handled. For example:

1: Request 14 samples:

The pump returns an index value of 1234 and the 14 samples of data.

2: Wait 1 second and request 14 samples

With no communications delay the pump would return an index of 1244 and the data. This means that the usable new data is $1244-1234=10$ samples of new data.

3: Wait 1 second and request 14 samples

This time there was a delay and a retry in the communications and the pump returned an index of 1256 and the 14 points of data. $1256-1244=12$ samples of new data and no loss of data.

Each sample can contain any of the available traces and is interleaved from lowest bit value to highest. Use 255 to get all 8 traces of data or 1 to get just Dispense Pressure.

Trace Bits

1 - Dispense Pressure

2 - Fill Pressure

4 - Dispense Position

8 - Fill Position

16 - Dispense Current

32 - System Status

64 - Valves Values

128 - Flow Sensor

Set Valves

This command allows changing the state of each of the internal valves independently. This is only allowed during the ready state.

The following bit values are used to set this 2 byte value representing the valve states:

NONE_VALVE 0

INLET_VALVE 1

ISOLATE_VALVE 2

BARRIER_VALVE 4

VENT_VALVE 8

PURGE_VALVE 0x10

OUTLET_VALVE 0x100

SVIDs

The SVID command allows the pump to be queried for specific data values in command. A list of System Variable ID values are sent as UINT16 data in the command. The command will return the requested data in the same order as it was requested. Note that different data types have different sizes (in bytes) as defined below:

'INT16' : 2
'UINT16' : 2,
'LONG' : 4,
'ULONG' : 4,
'FLOAT' : 4, IEEE 32 bit floating point
'STRING' : 2 bytes (USHORT) is the size of the returned string in
bytes followed the string of bytes.

The actual SVID values may change for different firmware revisions so the list is published with new firmware versions.